

**Adversity and Trauma:
Examining and Elaborating the Hybrid Model of
Stress Response as Applied to Belizean Contexts**

**by
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B.A. (Hons., Psychology), Simon Fraser University, 2022
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Thesis Submitted in Partial Fulfillment of the
Requirements for the Degree of
Master of Arts

in the
Department of Psychology
Faculty of Arts and Social Sciences

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SIMON FRASER UNIVERSITY
Spring 2025

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Abstract

Krupnik's (2019) Hybrid Model of Stress Response (HMSR) holds that for an adverse experience to be traumatic, it must: (i) be outside of normal life experiences and (ii) disrupt self-regulation. In settings like Belize, where violence such as daily gunfire is normalized, the concept of "normal" becomes problematic. The present study explores Belizeans' experiences of adversity through semi-structured interviews with 12 adults, analyzed via Interpretative Phenomenological Analysis (IPA).

Three major themes were identified: (i) long-term self-regulation disruption; (ii) adversity intensified by perceived failure of support; and (iii) coping and social support mitigating adversity severity. Building on these findings, drawing on inductive Grounded Theory principles, an Expanded HMSR (E-HMSR) is proposed, redefining "normal" as subjective normativity across three dimensions: Expectations & Experiences, Sociocultural Norms, and Moral/Ethical Judgment. The E-HMSR also introduces a dynamic view of self-regulatory dysfunction and proposes social support as a mediating factor in stress responses.

Keywords: IPA; adversity; trauma; stress responses; Belize; liberation psychology

Acknowledgments

First and foremost, I am deeply grateful to the research participants who contributed their time, energy, and, most importantly, their stories. Discussing trauma and adversity is no small ask, and their courage and generosity made this study possible.

I am deeply indebted to my supervisor, Dr. Robert G. Ley, whose unwavering support, patience, and dedication extended beyond the bounds of expectation. Despite his official retirement from Simon Fraser University, he continued to guide me through this process, taking time to review drafts and provide invaluable feedback.

I would also like to express my sincere appreciation to Dr. Sheri Fabian, whose qualitative methods course transformed my understanding of research. Before her class, I had little grasp of how qualitative methods worked, and her expertise and teaching laid the foundation for my methodological approach in this study.

A special thank you to Dr. Hali Kil for her steadfast guidance and encouragement from the very beginning to the end of this journey. Her insights, support, and kindness have been instrumental in shaping both this research and my academic growth.

I am also incredibly grateful to my family, friends, and colleagues who offered their time, thoughtful feedback, and encouragement throughout this process. Their insights and support helped refine my work and kept me motivated during challenging moments.

I couldn't have done this alone, and I'm truly thankful to everyone who helped me along the way.

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Introduction

Belize is a small tropical country, with a population of 397,483, and a land base of 23,000 square kilometers (Britannica, 2024b; Statistical Institute of Belize, 2024). Roughly, it is three quarters the size of Vancouver Island (Vancouver Island Economic Alliance, n.d.). Belize holds a unique identity as the only country that is both Central American and Caribbean (Britannica, 2024a), as well as being the only Central American country with English as its official language, a legacy of British colonialism that distinguishes it from its predominantly Spanish-speaking neighbors (The Consulate of Belize in LA, 2024). The diversity of Belize's population also reflects this distinct cultural identity: Mestizo/Hispanic/Latino communities make up 51.7%, Creole 25.2%, Mayan 9.8%, Garifuna 4%, and East Indian 1.5%, alongside smaller groups such as East Asians, Caucasians, and Mennonites (Statistical Institute of Belize, 2024). With the second-largest barrier reef in the world (e.g., the Belize Barrier Reef) and hundreds of sand cayes, mangrove forests, and coastal lagoons, Belize has become a popular tourist destination, with tourism accounting for about thirty percent of its economy GDP in 2022, as well as forty percent of employment (*Travel & Tourism Economic Impact 2023*, 2023; UNESCO World Heritage Convention, n.d.; Wunderman, 2024).

However, beneath the veneer of a tropical paradise, Belize faces significant domestic challenges, particularly given the high rates of adversity and trauma experienced by its population. One highly likely contributing factor is Belize's geographical position as a transit point for illicit activities through Central America, including human, weapons, and drug trafficking (Global Organized Crime Index, 2023). This location has likely exacerbated the country's struggles with organized crime and violence. According to the Government of Canada's (2024) travel advisories, visitors to Belize are advised to "exercise a high degree of caution" due to pervasive and severe levels of violent crime throughout the country.

Since 2018, the Belizean government has declared six States of Emergency (SoEs) to target gang and drug-related violence in high-risk areas – especially Southside Belize City (see: Mistler-Ferguson, 2021; Tzalam, 2023, 2024). These SoEs, typically limited to 30 days, have resulted in civic government measures such as raids on known gang territories, increased police presence on the streets, limits to public gatherings, and

a nightly curfew for minors (Leslie, 2024). The most recent SoE, declared in June 2024, expanded beyond Belize City – the largest urban area in the country – to include parts of the Cayo District (Tzalam, 2024). This SoE was subsequently extended for an additional 90 days following an attack on the Police Commissioner’s home (Love FM Belize News, 2024).

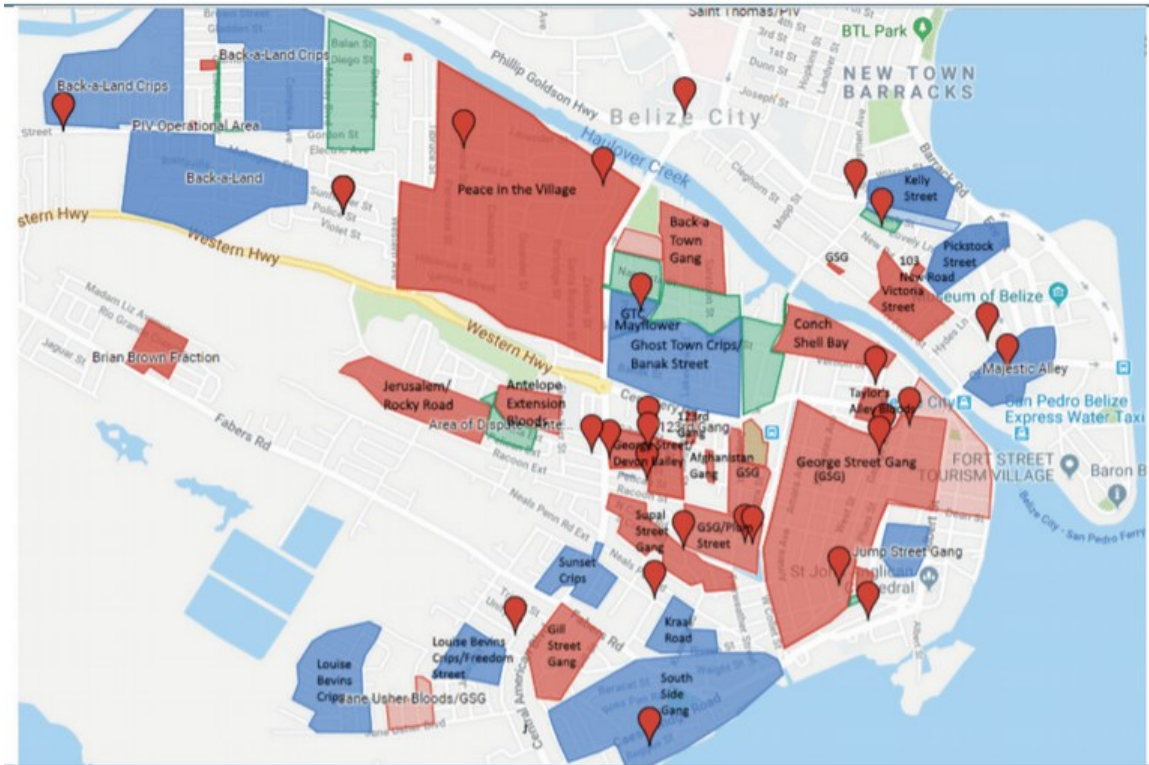
As illustrated by these SoEs, Belize, and particularly Belize City, struggles with frequent and severe adverse incidents, as well as prolonged periods of high stress and civilian danger. A recent online survey indicated that the prevalence of gang activity and shootings in Belize City has created a climate of fear and insecurity (Young, 2019). For instance, in a sample of gang members (N = 45, aged 16 to 41 years), 49% of respondents reported knowing of a person being shot on a daily basis. This pervasive threat of violence has significantly influenced daily life, with residents adopting behaviors to avoid victimization, such as limiting outdoor activities at night, avoiding certain neighborhoods, and enhancing home security. To provide context, and by way of comparison, in 2023, there were a total of 117 incidents involving firearms discharged with intent within British Columbia (Statistics Canada, 2023a). While 117 is a notable figure, a similar statistic is not available in Belize, perhaps due to the daily occurrence of gunfire, making it difficult to quantify. For Belizeans, this persistent threat to public safety is particularly concerning given its potential impact on mental health.

Beyond gang violence, Belize also faces high rates of domestic violence (reported at 304 individuals per 100k population), sexual violence (21 per 100k), and homicide (21 per 100k; Belize Crime Observatory, 2023). In contrast, Metro Vancouver, with a population 2.8 million, which is 10 times greater than Belize, recorded rates of homicide at 0.5 per 100k and sexual violence at 25 per 100k (Greater Vancouver Board of Trade, 2024; Vancouver Police Department, 2023). Additionally, Canada reported a domestic violence rate of 337 per 100k population (Statistics Canada, 2023b).

While domestic and sexual violence rates appear to be slightly higher in Canada and Vancouver, it is highly likely that the reported rates in Belize do not fully capture the true prevalence of these incidents. Given the small population and the large degree of gang influence, especially in Belize City, familial ties to these gangs may be common in certain areas, which may create significant barriers to reporting incidents (see: Figure 1 and 2). It is important to note that Figure 1 does not illustrate all gang territories,

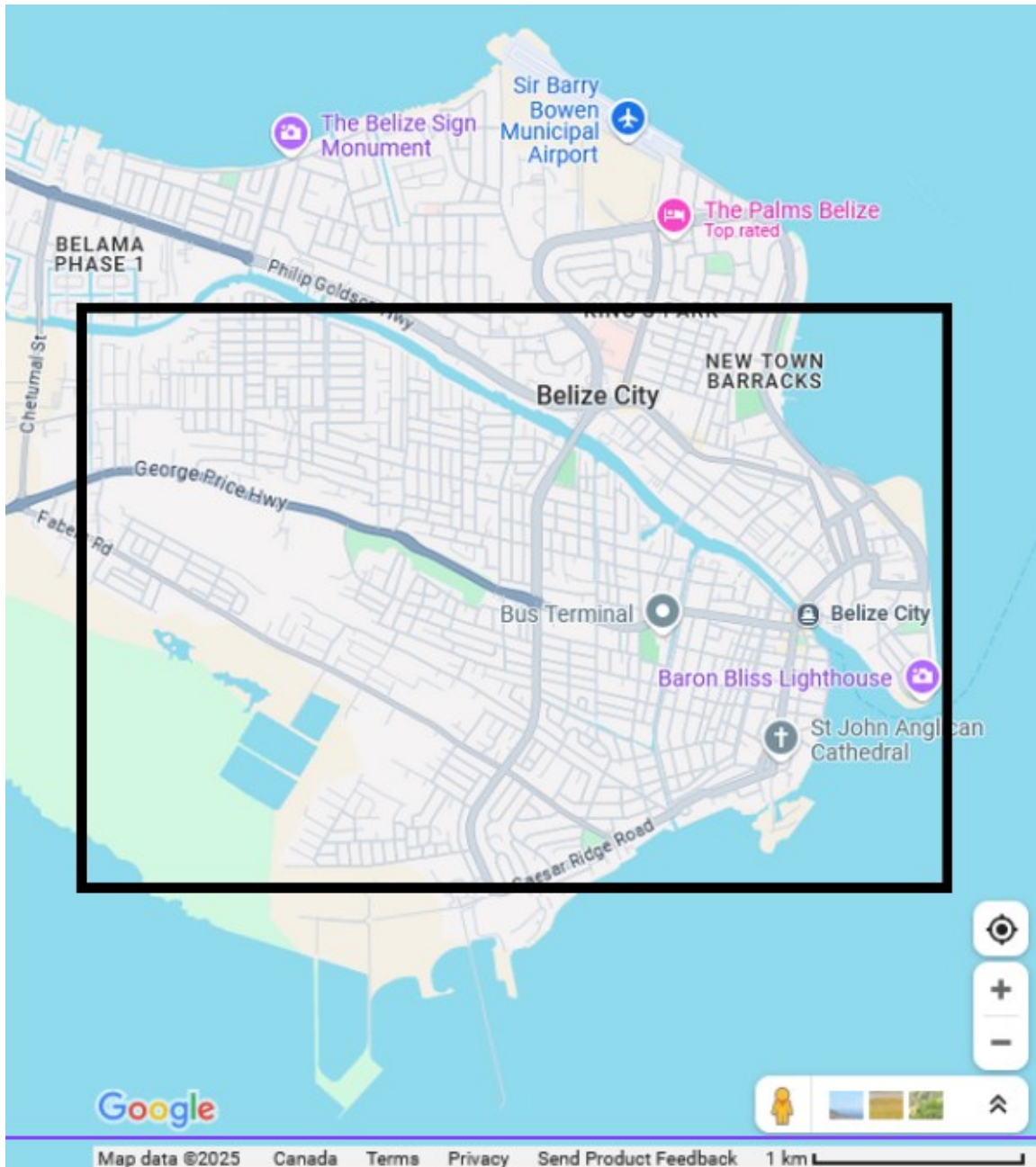
particularly in the Northside, where data gaps exist – I state this from firsthand experience, as I grew up in Belize City and continue to visit regularly.

Figure 1. Homicides and proximity to known gang territories, 2016–2017



Note: Annotated geospatial map of Belize City. Red dots indicate homicide locations, while shaded areas represent known active gang territories. Each gang territory is color-coded distinctly for differentiation. The map is divided by a major river, distinguishing Northside (above the river) and Southside (below the river) Belize City. Reprinted from Belize City Community Gang Assessment (p. 25), by M. Young, in D. M. King (Ed.), 2019, Inter-American Development Bank (IDB). Copyright 2019 by IDB. CC BY-NC-ND.

Figure 2. Belize City with Outlined Data Collection Area



Note: This map shows Belize City with a black outlined box marking the area analyzed in Young (2019). It provides additional context for understanding the extent of known gang territories across the entire city.

In this challenging environment, youths are exposed to extremely high levels of traumatic experiences linked to violence, including exposure to, or experiences of severe physical harm or death (see Table 1; Gayle et al., 2010). The widespread concerns about safety are reflected in Young's (2019) Community Gang Assessment report, which

revealed that 84% of parents (N = 303, random sampling on residents of Belize City, face-to-face surveys) believed their child could be harmed by gang violence. A community survey (N = 567, randomly selected youth aged 12 – 22, Belize City) revealed that one in five youth had current or prior gang involvement (Young, 2019). Given the prevalence of gang membership, exposure to violence is a frequent reality for many. 43% of youth (N = 159, ages 12 – 22, non-random sampling due to poor coverage in some Belize City areas, face-to-face and online surveys) felt that they or a friend were likely to be injured as a result of gang violence. In such a volatile and threatening environment, harsh parenting practices, including physical punishment and yelling (Lie-A-Ling et al., 2018), likely further contribute to the adversity faced by Belizean children and teenagers.

Table 1. Assessment of Total Multiple Trauma Experienced by Gender

Totality of Experiencing Trauma Personal + Witness + Know of Persons	Gender			Total
	Male	Female	Not Stated	
% of youth with absolutely no trauma experienced	1.7%	0.0%	0.0%	0.8%
% of youth with 1 - 2 experiences in total	2.3%	2.9%	0.0%	2.5%
% of youth with 3 - 5 experiences in total	1.7%	9.7%	0.0%	5.7%
% of youth with 6 - 10 experiences in total	11.4%	15.4%	66.7%	13.9%
% of youth with more than 10 experiences in total	82.9%	72.0%	33.3%	77.1%
TOTAL	100.0%	100.0%	100.0%	100.0%

Note: n = 175 male and n = 178 female students aged 12-21 were randomly selected from 19 secondary schools (17 in urban Belize and Cayo, 2 in rural areas). A quantitative survey was administered by a guidance counselor or teacher to gather data on students' exposure to and experiences with violence, including stabbings, shootings, beatings, and police abuse. Respondents reported on the number of people they know or have witnessed affected by these events, as well as their own personal experiences. Reprint from Gayle et al. (2010)

Adversity in Belize also extends beyond violence to encompass systematic inequalities – deeply ingrained disparities in access to resources and opportunities based on demographic factors. Economic hardship is a critical issue, with 35.7% of the population living in poverty, according to the 2022 Census (Statistical Institute of Belize, 2024). This level of impoverishment is a stark contrast to Canada, where the national poverty rate is significantly lower at 8.1%, with Vancouver at 11.2%, as found in the 2021 Census (Government of Canada, 2022b). In addition to economic challenges, colorism and racism (Whittaker-Augustine, 2021), as well as discrimination based on sexual

orientation (Beck et al., 2017), disproportionately affect marginalized groups in Belize. For example, the Government of Canada (2024) advises queer visitors to Belize to avoid public displays of affection due to the risk of harassment and verbal or physical abuse.

In addition to these systemic inequalities, family instability may be a significant source of adversity in Belize. While 16.4% of families in Canada are single-parent households, the prevalence is substantially higher in Belize, where 45.9% of adolescents in a nationwide sample of 676 live in such households (Government of Canada, 2022a; Vairez et al., 2022). In Belize City, the majority of youth face some form of family disruption, with 71% reporting that they either live in a single-parent household or apart from their biological parents; among gang-involved youth, this figure is even higher at 89% (Young, 2019). Here, youths from single-parent households are more likely to report experiencing more risk factors for gang-involvement, such as being suspended or expelled from school for behavioural issues, expressing antisocial beliefs, and regularly engaging in risky or dangerous behaviour. This high rate of family instability in Belize may have serious implications regarding exposure rates to adversity and/or trauma.

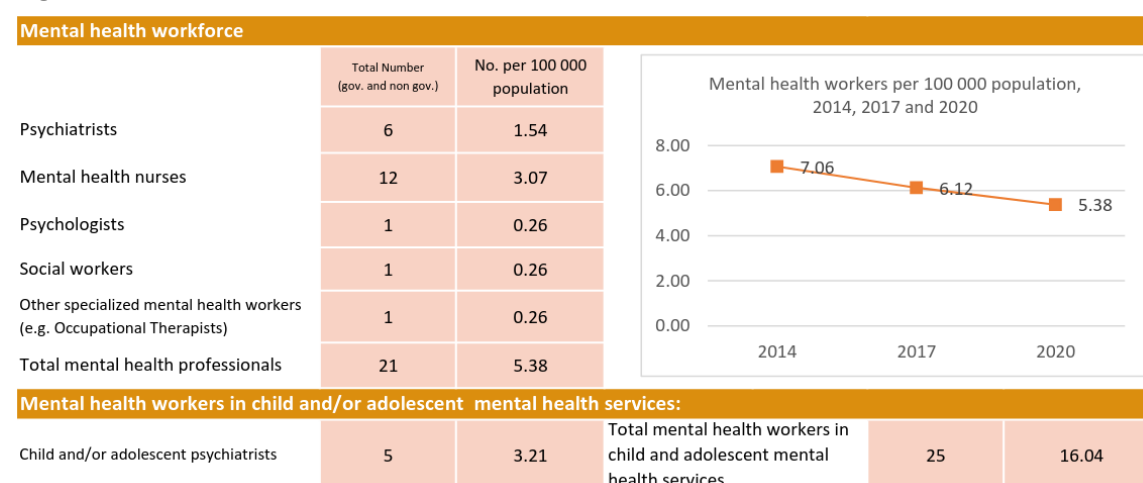
According to Dorrington et al.'s (2019) cross-sectional population study (n = 3995 twins, n = 2019 singletons, Sri Lankan residents), childhood experiences of maladaptive family functioning are not only strongly associated with increased trauma exposure but also with increased vulnerability to PTSD and other DSM diagnoses. Importantly, this relationship remained significant even after accounting for other factors like exposure to interpersonal violence, cumulative trauma, and other psychiatric disorders. In the context of Belize, where family disruption is widespread, these findings suggest that the impact on mental health and trauma exposure may be particularly severe.

Despite the high rates of adversity, trauma, and risk of trauma exposure in Belize, there is a total absence of accurate statistical data on mental health within the Belizean population (Ministry of Health and Wellness, 2023). The lack of epidemiological studies makes it challenging to accurately assess mental health outcomes within this high-adversity socio-cultural context. Although Gayle et al.'s (2010) large scale research project explored issues of adversity and trauma in Belize (e.g., see Table 1), their study, which was conducted over 15 years ago, focused primarily on the criminological aspects of social violence and its ecology within the country, rather than on psychological aspects of its impact. Existing psychological data, primarily derived from the Belize Health

Information System (BHIS), indicates that anxiety disorders (45%), affective disorders (22%), and schizophrenia-related disorders (8%) were the most prevalent mental health conditions recorded from 2018 to 2021 (Ministry of Health and Wellness, 2023). However, these data are very limited and do not provide a comprehensive picture of mental health across the population.

There are significant systemic barriers in mental health care and research within Belize, including the lack of mental health legislation and a shortage of services (Ministry of Health and Wellness, 2023). For instance, in light of the trauma rates found in Table 1, it is concerning that there are no inpatient or outpatient mental health facilities specifically for children and adolescents (World Health Organization, 2022). Additionally, no programs exist to support early childhood development or promote and prevent paternal/maternal mental health challenges. Belize also faces a critical shortage of mental health workers (see Figure 3).

Figure 3. Mental Health Atlas 2020: Mental Health Worker Statistics in Belize



Note: From Mental Health Atlas 2020 Country Profile: Belize (p. 2), 2022, World Health Organization (WHO). Copyright 2022 by WHO.

These significant systematic barriers stem from the widespread stigma and discrimination surrounding mental illness in Belize (Ministry of Health and Wellness, 2023). Negative attitudes toward mental health – present at all levels of society, including professional and political domains – hinder efforts to: (i) prioritize mental health, (ii) integrate services into the broader healthcare system, and (iii) advocate for evidence-based treatments. Such stigmatization not only discourages individuals from seeking help but also undermines efforts to create a more supportive environment for mental health care.

Compounding these systemic barriers to mental health care is the significant gap in psychological research on trauma and adversity in the country, which hinders the development of culturally and contextually relevant infrastructure and interventions. Without data to inform these efforts, it becomes nearly impossible to design effective programs that address the specific needs of the Belizean population. Notably, the only psychology-oriented study addressing trauma in Belize is a dissertation published on ResearchGate. This dissertation, which involved in-depth narrative interviews with 11 girls aged 13 to 18 years, found that childhood trauma (including witnessing violence at home, community violence, and sexual abuse) played a significant role in female youth gang involvement (DeGuerre, 2023).

Although much of the present discussion on adversity and trauma in Belize centers on gang-related violence, this present emphasis reflects the availability and accessibility of data rather than the full scope of adversity and trauma in the country. Belize's geographical position, lack of infrastructure, and stigma surrounding mental health all contribute to this narrow focus, alongside the visible and pressing impact of gang-related violence on public safety, as seen in the frequency of State of Emergency decrees. While this context helps to explain the emphasis of existing data on gang-related violence in existing literature, it also highlights significant gaps in the understanding of other forms of adversity, such as discrimination and natural disasters. Despite the lack of research and documentation on non-gang-related adversities, they may be equally pervasive and harmful to the population. To address these pressing gaps, this study aims to explore the broader experiences of adversity and trauma within Belizean contexts, contributing essential insights to inform culturally relevant mental health policies and practices.

Literature Review

Understanding the relationship between adversity and trauma in a Belizean context requires a thorough exploration of the foundational concepts related to these phenomena. This necessity involves reviewing definitions, theories, and frameworks related to adversity, trauma, and stress, alongside the socio-cultural factors that influence their expression and impact. Although trauma research from other countries may offer some guidance, applying these frameworks to Belize demands careful attention due to the nation's unique sociocultural contexts and realities.

Adversity and Trauma

The conceptualization of adversity and trauma presents significant challenges in psychological research and clinical practice, particularly when considering the wide spectrum of adverse experiences that may be traumatizing (Krupnik, 2019). While all trauma can be considered a form of adversity, not all adversity is viewed as trauma – a distinction that complicates the definition of trauma and remains a topic of debate. Two contrasting perspectives dominate trauma literature.

One perspective, categorical in nature, defines trauma as a distinct and narrowly specific phenomenon usually associated with post-traumatic stress disorder (PTSD; Krupnik, 2019). This viewpoint features specific diagnostic criteria, such as those outlined in the DSM-5-TR, where trauma is defined exclusively as the following: exposure to death, injury, or sexual violence through direct experience, witnessing the event, learning it happened to a close person, or repeated extreme exposure to event details (American Psychiatric Association, 2022b).

To clarify, in this DSM-5-TR framework, trauma is not synonymous with PTSD. Instead, trauma refers to the distressing or harmful experiences that can potentially lead to the development of PTSD symptoms. While PTSD is a clinical diagnosis characterized by specific criteria, such as intrusive thoughts, avoidance behaviors, and hyperarousal, trauma represents the events or circumstances that may trigger these symptoms. This distinction is important as not all individuals who experience trauma will develop PTSD. For instance, in a nation-wide Canadian survey, while 63% of adults

reported exposure to at least one traumatic event, only 8% reported moderate to severe PTSD symptomatology (Statistics Canada, 2024).

However, the “DSM-based” categorical approach to trauma is fundamentally flawed for several reasons and has many critics, despite its popularity and commonality in mental health fields and psychopathology diagnoses in North America. Within the PTSD criteria, the terms 'trauma' and 'traumatic events' are used interchangeably, leading to conceptual confusion. Additionally, the DSM-5-TR has been criticized for excluding many forms of adversity, such as psychological harm, e.g., prolonged emotional abuse, that can result in PTSD symptoms (Dalenberg et al., 2017). While there is some acknowledgment of trauma from cumulative experiences, the DSM does not sufficiently encompass the full range of chronic adversities that many individuals face, particularly in environments like Belize, where ongoing and pervasive adverse conditions can lead to significant trauma responses, including PTSD. For example, chronic stressors associated with PTSD, such as severe economic instability (Holmes et al., 2021) and racial discrimination (Kirkinis et al., 2021), are not recognized within the DSM framework as trauma-inducing events.

On the other hand, in trauma literature, a second and alternate perspective on trauma significantly widens the concept, as it rests upon broad definitions such as “any event that has had a lasting negative effect upon self and psyche” (Krupnik, 2019, p. 1). Thus, instead of utilizing rigid event-based criteria, this perspective frames trauma and adversity as interconnected experiences that exist on a continuum. For example, a person may experience ongoing exposure to poverty or racial discrimination – each of which may not be classified as trauma in a traditional sense. However, subjectively, some individuals may perceive these experiences as traumatic.

While this second alternative approach broadens the definition of trauma to include a wide range of negative life events – subjectively determined as traumatic or otherwise adverse – it may risk oversimplifying complex experiences of human suffering (Krupnik, 2019). For example, experiencing a natural disaster and enduring ongoing emotional abuse both constitute significant negative events. However, the psychological outcomes of these experiences can be very different, in terms of externalizing and/or internalizing behaviours. Furthermore, by conceptualizing all adverse experiences as potentially traumatic, there is a risk of viewing emotional and psychological responses as

indicators of trauma, rather than as adaptive or appropriate, albeit challenging, human experiences. This perspective can lead to an overemphasis on trauma as the primary lens through which all adverse experiences are understood.

This continuum-based approach might also unintentionally minimize or invalidate the trauma experienced by those whose situations are perceived as less severe (Krupnik, 2019). Individuals who experience less severe adverse events may feel their experiences are less important when compared to more severe traumas on the continuum. Additionally, comparing different forms of suffering is both ethically and conceptually challenging. What is devastating for one person may not be perceived the same way by another. Comparisons raise questions about the validity of each individual's psychic pain; both are valid and deserve acknowledgment, regardless of their differing nature. Thus, it is essential to approach each person's experience with non-judgementality, openness, and empathy, while recognizing the unique contexts and personal perceptions that shape how trauma is felt and processed by different individuals.

To address these limitations, the *Hybrid Model of Stress Response* (Krupnik, 2019) focuses on the individual's response to the stressful events rather than solely upon the events themselves. According to Krupnik's model, individuals with "robust self-regulation" and related effective coping abilities would be less likely to experience traumatization. Krupnik (2019) suggests that the threshold for self-regulatory dysfunction could be identified through symptom clusters found within PTSD and stress-induced pathology, e.g. intrusive re-experiencing, stress-specific hyperarousal, avoidance, and emotional numbness. The Krupnik model combines continuum-based, dimensional, and categorical views, acknowledging that stress and adversity exist on a spectrum while also recognizing that extreme stress can transition into the category of "trauma" when it overwhelms a person's ability to cope. Three types of stress responses are described: normative, pathogenic, and traumatic (see Table 2).

Table 2. Criteria for Each Stress Response Classification Based on Krupnik's Model

Stress Responses	Outside of Normal Life Experiences	Breakdown of Self-Regulatory Functions
Normative		
Pathogenic		X
Traumatic	X	X

Krupnik (2019) argues that for an experience to be considered truly traumatic, it must be both outside of normal life experiences and result in a breakdown of self-regulatory functions. This threshold differentiates between “normal” adversities that might cause mental health issues (pathogenic stress responses, such as tension and anxiety) and extraordinary events that significantly disrupt a person's ability to manage stress (traumatic stress responses, such as PTSD). The definition of what constitutes “normal” is subjective and varies significantly across individuals, shaped by their backgrounds, cultural contexts, and personal experiences.

However, in regions like Southside Belize City, where violence and adversity have become normalized, Krupnik's (2019) model presents unique challenges in differentiating between pathogenic and traumatic stress. Adversities that might be considered extraordinary in other countries, such as daily shootings, are statistically common within this geographical and sociocultural context. Even for individuals who have not directly witnessed or experienced these events, the consistent presence of shootings in the news and the shared experiences of peers or close family members who have been victimized reinforce this normalization.

It is important to note that this process of normalization would not diminish the potential for such stressors to be experienced as traumatic. The blurring of the boundary between pathogenic and traumatic stress is further complicated by the lack of empirical data on how Belizeans perceive and experience these sustained adversities, particularly regarding the psychological impact of prolonged exposure to interpersonal threats or frequent instances of violence. For example, in some neighborhoods, fireworks are often used to celebrate holidays or as weekend entertainment, but due to the normalization of shootings, many people frequently mistake these sounds for gunfire. Some youths even

set off large amounts of fireworks around holidays to mimic the soundscape of a “war zone”, reflecting how deeply ingrained the culture of violence has become in their daily lives (“Ambergris Caye Holiday ‘War Games,’” 2019).

According to Myers et al. (2015), lifetime cumulative adversities – such as experiences of discrimination, childhood sexual abuse, childhood family adversities, and chronic stresses – are strongly predictive of psychological distress and dysfunction. Thus, growing up and living in a persistent state of potential danger, albeit “normalized,” may have unique psychological impacts that differ from those observed in post-trauma contexts where the threat has passed. At what point does the adversity become traumatic, if the adversity is normalized in Belize? The relationship between adversity and trauma may not be linear or universal, especially within a diverse socio-cultural context.

Stress Theories

Krupnik’s *Hybrid Model of Stress Response* (HMSR) fundamentally views trauma through the lens of an individual’s capacity to regulate stress, making it essential to review theories of stress to contextualize this model. While Krupnik did not cite existing stress theories in their original exposition of their model, they were likely influenced by Hans Selye’s pioneering work and subsequent stress models that have evolved since then.

The term “stress” was first coined by Selye to refer to a “nonspecific response of the body to any demand upon it” (Selye, 1936, as cited in Lu et al., 2021). This definition of stress became more specific when layered upon Cannon’s (1935, as cited in Lu et al., 2021) concept of homeostasis – the body maintaining a stable internal state despite external influences. Thus, stress was a physiological concept defined as a state in which the body’s hypothalamic-pituitary-adrenal (HPA) axis and sympathoadrenal systems activate in response to threats that disrupt homeostasis. In simpler terms, the definition of stress was as follows: when the body’s ability to maintain a stable internal environment is challenged. However, this foundational definition has evolved significantly, with the concept of stress becoming more interdisciplinary to encompass psychological and environmental factors. Researchers now recognize that stress can be understood through various theoretical frameworks, all of which can be categorized into generally three

perspectives and three classifications (Schwarzer & Luszczynska, 2013; Shahsavarani et al., 2015).

Perspectives to Understanding “Stress”

Theories of stress can be categorized into three main perspectives: (i) response-based, (ii) stimulus-based, and (iii) cognitive-transactional-based. While Krupnik’s HMSR does not explicitly draw on these foundational theories, its conceptualization of stress and trauma shares similarities with aspects of each perspective.

Response-based

General Adaptation Syndrome (GAS; Selye, 1956) is one of the foundational theories of stress and offers a model for understanding how individuals respond physiologically to prolonged stress. It describes the body's physiological response to stress in three stages: alarm, resistance, and exhaustion. The alarm stage is the initial response to a stressor, often referred to as the “fight-or-flight” response. The body reacts with an increase in heart rate, blood pressure, and the release of stress hormones like cortisol and adrenaline. If the stressor persists, the body attempts to adapt to the demands; in other words, the body attempts to resist. Prolonged resistance to the stressor can lead to wear and tear on the body, manifesting as diseases of adaptation, such as hypertension. If the stressor continues for an extended period without adequate recovery, the body’s resources are depleted, leading to exhaustion and potential harm to the body, such as illness or death.

However, since GAS is a physiological model, it does not account for the broader psychological or social factors involved in stress. Despite this limitation, a comparison still can be made with the HMSR, as both models recognize a critical threshold where excessive stress negatively impacts health. While GAS defines this threshold as the point at which the body’s physiological capacity to resist stress is overwhelmed, leading to physical harm and damage, Krupnik’s model identifies it as the threshold where self-regulatory mechanisms break down, resulting in psychological harm, such as psychopathology caused by pathogenic or traumatic stress.

Stimulus-based

The stimulus-based perspective on stress focuses on the external demands placed on individuals, i.e., the stressors, and their specific characteristics, such as whether it is physical, social, or task-related (Lyon, 2012). According to this view, certain life events or stressors impose unique challenges on an individual's resources and coping capacity, which can lead to stress. This perspective emerged from the work of Holmes and Rahe (1967), who introduced the Social Readjustment Rating Scale (SRRS). The SRRS sought to measure the intensity of life events by assigning a numerical value to each event, based on how much coping effort each adverse life event was believed to require. The more effort someone needs to adjust to a life event, the more "severe" that event is considered. However, this approach assumes that all individuals experience the same level of stress from a given event, regardless of whether the event is perceived as positive or negative. This consideration overlooks the subjective nature of stress and fails to consider how different individuals' coping abilities and resources influence their experience of a stressor. For example, while the loss of a job may be perceived as highly stressful by some, others might see it as an opportunity for growth or a career change, or simply as a welcome and benign event.

A similar approach and set of corresponding limitations can be observed in commonly used trauma assessments like the Trauma History Questionnaire (THQ; Hooper et al., 2011). The THQ categorizes traumatic events into predefined areas, such as crime-related incidents, disasters, or unwanted physical and sexual experiences. It primarily emphasizes the occurrence, frequency, and context of specific events (e.g., "Have you experienced [event]?" and "How many times did it happen?") and does not assess the individual's emotional response to the events or subjective perception of their impact. Similarly to the stimulus-based perspective, this categorical approach to understanding trauma assumes that the nature of an event inherently determines the severity of stress experienced, and thus, whether it is categorized as traumatic. In contrast, Krupnik's HMSR avoids this problematic assumption by shifting the focus from the objective characteristics of events to the individual's subjective response to stress. By prioritizing how people perceive and react to stressors, Krupnik's model recognizes that trauma is not solely determined by the event itself, but also by the unique ways in which individuals interpret and process their experiences.

Cognitive-transactional-based

Unlike the response- and stimulus-based perspectives, the cognitive-transactional perspective on stress emphasizes the dynamic interaction between the individual and their environment (Schwarzer & Luszczynska, 2013). This approach, developed by Lazarus (1966, 1991, 2006, as cited in Schwarzer & Luszczynska, 2013), views stress not as a fixed attribute of a situation or a physiological response, but as a process arising from the relationship between external demands and the individual's perceptions and coping resources. In this framework, stress arises when individuals perceive that the demands of a situation surpass their ability to manage them and pose a risk to their well-being. Additionally, within this model, stress is a process shaped by personal and environmental factors, with coping and appraisals mediating its immediate emotional and physiological effects, as well as long-term impacts on health and well-being.

Lazarus' model identifies two key processes: primary and secondary appraisal (Schwarzer & Luszczynska, 2013). During primary appraisal, individuals evaluate whether a situation is irrelevant, benign, or stressful. If deemed stressful, secondary appraisal follows, where individuals assess their ability to manage or cope with the demands. These appraisals are influenced by personal variables, such as beliefs or commitments, and environmental factors, such as situational constraints.

Unlike the stimulus-based perspective, which assumes that certain events are universally stressful, the cognitive-transactional perspective recognizes that the same event can be experienced differently depending on personal, cultural, and contextual factors. Similarly, Krupnik's HMSR shares this emphasis on the subjective experience of stress and individual variability in coping and responses. Building on these principles, Krupnik's model further describes how overwhelming stress disrupts an individual's capacity for regulation, drawing a threshold at which stress transitions into trauma.

Classifications of “Stress”

Overall, according to Shahsavarani et al. (2015), the understanding of stress can be categorized based on three types of classifications: (i) the nature of the stressor, (ii) its influence on the individual, and (iii) the duration of exposure. Stressors can be psychological (emotional, cognitive, or both) or physiological (such as mitochondrial stress). Also, stress can have differing impacts: it can be positive, referred to as *eustress*, which

enhances adaptive capacity, or negative, known as *distress*, which challenges homeostasis and impairs health. A third form, *sustress*, refers to insufficient stress that limits an individual's capacity to adapt, potentially jeopardizing health. Stress can also vary in duration. Acute stress arises from short-term challenges, while chronic stress results from prolonged exposure to stressors.

Krupnik's HMSR closely aligns with the second classification of stress by focusing on the individual's response to stress and categorizing it into three distinct types: normative, pathogenic, and traumatic. In the Krupnik framework, *eustress* can be mapped to the normative stress response. *Distress* can be mapped to the traumatic stress response, as it challenges self-regulation and can lead to long-term harm, such as PTSD. *Sustress* would correspond to pathogenic stress. These comparisons not only help clarify Krupnik's model but also expand upon the specifications of each stress response, particularly in the context of the normative and pathogenic categories. While Krupnik's model offers valuable insight, it remains in its preliminary stages and lacks detailed elaboration of these two categories.

In the case of pathogenic stress, Krupnik's model suggests that responses like anxiety could fall within this category, as they reflect a disruption in self-regulation without necessarily escalating into a traumatic response. Prior to making the above comparisons, it was unclear how pathogenic stress would distinctly differ from traumatic stress, especially considering that both can result from disruptions to self-regulation. This clarification is particularly relevant in contexts like Belize City, where the definition of "normative" experiences may be blurred by pervasive adversity, such as violence and poverty.

However, in light of the third classification of stress as described by Shahsavari et al. (2015), a key limitation of Krupnik's model is that it does not account for the temporal dimension of stress exposure. Specifically, the HMSR does not fully address how an individual's capacity for self-regulation may fluctuate over time in response to a single sustained stressor. For instance, the nature of the stress response may change: a prolonged exposure to a stressor can diminish the individual's ability to cope, potentially transforming what was initially a normative stress response into something more harmful, such as one that is pathogenic or even traumatic.

Bridging Stress Theories and Trauma Research

While the foregoing review of stress theories provides important context for understanding and potentially expanding upon the *Hybrid Model of Stress Response*, the review also highlights a critical limitation: these stress theories do not specifically address trauma or the transition from stress to trauma. This shortcoming reinforces the value of Krupnik's model, as Krupnik explicitly addresses the threshold at which stress transitions into trauma - a crucial consideration for understanding traumatic experiences in Belize's high-adversity context.

Cultural Considerations

Belize's ethnolinguistic diversity and cultural foundations may significantly shape how individuals experience and perceive trauma. Socio-cultural elements can act as protective factors against trauma or contribute to the risk of developing PTSD and other trauma-related conditions, such as anxiety, depression, or substance use disorders (Ford et al., 2015). For example, strict cultural norms or expectations might lead individuals to suppress or ignore emotional distress, which could manifest in culturally specific ways such as emotional withdrawal, reluctance to seek mental health care, or the manifestation of physical symptoms of stress. Additionally, factors such as exposure to neighborhood risk, early childhood adversity, and prolonged adversity, can interact with stress responses at a physiological level (Doan & Evans, 2017).

One of the most influential cultural factors in Belize may be religiosity, given the strong Church-State partnership. This religious presence is especially evident in public education settings: Christian churches co-manage a large proportion of schools (60% of primary schools, 40% of secondary, 50% of tertiary), and weekly "spirituality" classes are mandatory (Office of International Religious Freedom, 2023). In 2022, 49.7% of Belizeans identified as part of Christian denominations such as Roman Catholic, Pentecostal, Seventh-Day Adventist, and Anglican (Statistical Institute of Belize, 2024). For many Belizeans, this involvement in religious life may help reduce the likelihood of experiencing traumatic stress responses. For instance, religious beliefs, such as attributing stressors to a divine plan or relying on God's support, often serve as powerful coping mechanisms, particularly during periods of acute stress, like bereavement (Graça & Brandão, 2024). Furthermore, religious communities tend to offer strong social support networks,

with relationships often characterized by higher levels of trust and reciprocity compared to general social circles (Shapiro, 2022).

However, religious practices can also have negative effects on mental health. Negative religious coping strategies – such as interpreting hardships as evidence of abandonment by God – have also been linked to increased emotional distress, impaired psychological recovery, and poorer well-being (Graça & Brandão, 2024). Moreover, the well-documented prevalence of sexual abuse within the Roman Catholic Church (for instance, see: Dressing et al., 2017; Lecaros & Suárez, 2024) raises concerns about trauma exposure in certain Belizean communities. Beyond the direct harm to survivors, clergy abuse erodes trust in religious institutions, weakening community support systems and exacerbating long-term psychosocial distress.

Religious institutions have also played a significant role in shaping policies and reinforcing norms that negatively impact many demographic groups. In 2023, the Belize Council of Churches (BCC) opposed an anti-discrimination bill, preventing it from reaching parliament (Office of International Religious Freedom, 2023). While the bill aimed to protect women, victims of violence, and other vulnerable groups, the BCC opposed it due to its inclusion of protections for same-sex partners and an expanded legal definition of family. Similarly, despite the decriminalization of “sodomy” in 2016, religio-political challenges to queer rights continue to persist (Guest Contributors, 2017; Lavers, 2020).

Extensive research has shown that discrimination and marginalization – whether based on gender, age, or sexual identity – have widespread and long-term negative impacts on mental health (for instance, see: Grigoreva & Szaszko, 2024; Lei et al., 2021). In Belize’s conservative religious environment, women and girls, often considered marginalized groups, face restricted autonomy due to structural sexism and gender discrimination (Funa, 2023). Research shows that women in sexist religious environments report poorer mental health outcomes (Homan & Burdette, 2021). These patterns suggest that the mental health benefits of religious communities may not apply to those excluded from power and status within their communities.

Unfortunately, there is a lack of relevant research that addresses how this environment affects men and boys. However, evidence suggests that vulnerable individuals within this demographic may turn to gang involvement as a means of coping, for similar

benefits that could be found in religion, such as respect, loyalty, social support networks, and emotional support. A survey of gang members in Belize City (N = 45, predominantly male, aged 21 years+, having spent half their lives in gangs) found that a majority experienced these positive aspects of gang involvement, in addition to protection, safety, and financial stability (Young, 2019).

While these socio-cultural factors provide valuable insights into the experiences of adversity and trauma in Belize, there is little psychological research addressing how Belizeans perceive and interpret these adverse phenomena. Understanding cultural norms, religious practices, and the ways individuals express and cope with distress is essential to contextualizing trauma within Belize's unique social and cultural framework.

Research Objectives

The present study aims to not only advance theoretical knowledge of trauma but also create a more culturally relevant framework for describing trauma in Belize, a diverse under-represented population in research. To achieve this, this study has two main objectives.

The first objective is to explore the experiences of adversity and trauma among Belizeans, focusing on the types of adversities they encounter and how they perceive and interpret these experiences. Given Belize's significant cultural, ethnic, and social diversity, including a diverse range of perspectives in this study is essential to fully reflect the experiences of its various populations. By exploring these perspectives, this study addresses the critical scarcity of psychological research in the national context and may inform the development of culturally relevant mental health interventions, services, policies, and legislation, all of which are currently lacking.

The second objective is to expand upon Krupnik's (2019) *Hybrid Model of Stress Response* (HMSR) and propose an adapted version that is more culturally relevant to Belizean contexts. While the HMSR suggests that trauma arises from experiences that are outside the norm, in Belize, severe adversities are often normalized, raising questions about the model's applicability. Rather than categorizing Belizean experiences as traumatic or non-traumatic based on the HMSR to assess the model's fit, this study aims to modify the HMSR by building upon insights from Belizean experiences. These modifications include adjusting the "normal" criterion and expanding the self-regulation condition to better reflect the dynamic nature of stress responses.

Positionality Statement

I was born into an upper-middle-class family in North-side Belize City, where I grew up in a blend of Taiwanese heritage from my parents and Belize's diverse cultural environment. Although I lived in a relatively safe neighborhood, there have been gang-related incidents.

I first attended St. Joseph Roman Catholic Primary School, a low-income institution where issues such as alcohol consumption and authoritarian disciplinary methods, including corporal punishment, were prevalent. Peers, as young as eight years old, were suspended for drinking rum at school. Later, I attended Saint Catherine Academy, a high-income, all-girls Catholic high school where students were – and continue to be – expelled based on suspicions of being gay or for becoming pregnant. Despite the differences in socioeconomic backgrounds, students in both schools shared similar struggles, such as the loss of peers to gang violence and suicide.

In my research, I occupy a complex position as both an insider and an outsider. Having been educated and socialized in Belize until I was 18 years old, I share a cultural background with my participants. However, being primarily engaged in the niche music community in Belize puts me in an outsider's role as well. I have had a relatively sheltered experience, compared to most peers in Belize.

In Belize City, I had come to accept the prevalence of violence, including gang activity, as a normal part of daily life. It was not until I began living in Vancouver in 2018, in a city with relatively low crime rates and more structured support systems, that I began to comprehend the extent of Belize's challenges in these areas. For example, while I feel safe walking alone at night in Downtown East Side Vancouver, I've noticed that many of my Canadian colleagues express fear about doing the same.

My academic journey has further shaped my perspectives within research. Educated in psychology in both Belize and Canada, I have been influenced by diverse cultural and academic environments. This has led me to personally and philosophically embrace a constructivist perspective: (i) "realities" are multiple and constructed by individuals based on their experiences and interactions; and (ii) knowledge is co-constructed in the interaction of the researcher's interpretations and the interviewees' perspectives.

I am aware that my positionality comes with biases, assumptions, and privileges. Growing up in Belize and now living in Canada, I may carry cultural assumptions that no longer fully reflect the current realities in Belize. I recognize the risk of viewing Belizean issues through a Canadian lens, influenced by my experiences in a country with greater resources and relative stability. Additionally, my educational and socioeconomic privileges, such as access to higher education and academic resources in Canada, provide me with certain advantages that my participants might not have. This disparity requires me to be particularly mindful of power dynamics in the research process.

To address these concerns, I engaged in reflexive practices, regularly reflecting on how my background shapes my research. During data collection, I actively sought feedback from participants to ensure their voices are authentically represented.

Methods

Qualitative data from a larger study conducted by Joy Lee-Shi, which examined adversity, trauma, coping strategies, available support, and post-traumatic changes among Belizean adults, was analyzed. The original study was approved by the Simon Fraser University Research Ethics Board (REB). Since the current project only involves secondary analysis of anonymous data from the approved study, it was exempt from further REB review under TCPS Article 2.4.

Participants

Belizean adults (N = 12) were recruited using a combination of snowball and purposive sampling to ensure diversity across age, gender identity, and sexual orientation. Inclusion criteria required participants to be Belizean, 18 or older, fluent in English, and willing to discuss personal experiences. Recruitment began through convenience sampling within existing personal networks, with potential participants being contacted directly through social media platforms or email. Initial contact involved a brief study description and an inquiry about participation interest. Three participants were close friends and also contributed to the recruitment process by referring additional individuals.

As recruitment continued, snowball sampling became the primary method, with participants and individuals in personal networks referring friends and acquaintances who met the study criteria. Midway through recruitment, when queer participants formed the majority of the sample, efforts shifted toward purposive sampling to include more straight, cisgender individuals. Individuals who expressed interest received a formal consent form (See: Appendix A) outlining the study's purpose, details, and participation instructions.

Data Collection

A semi-structured interview protocol (See: Appendix B) was developed and implemented for data collection. Individual interviews were conducted by the researcher via Zoom, and five participants chose to keep their cameras off due to preferred personal comfort or bandwidth limitations. The interviewer confirmed this preference with each

participant and also kept their camera off for one participant due to bandwidth issues, while using video for all other interviews. The participants were informed that the interviews would last about one hour or until they felt that they had fully shared their experiences. At the one-hour mark, the interviewer notified participants of the time and offered the option to conclude the interview. The majority of participants opted to continue their interview past the one-hour mark, resulting in one hour to two and one-half hour long interviews.

Throughout the interviews, when appropriate, the interviewer shared brief personal observations about life in Belize. These self-disclosures, always relevant to the topic at hand, helped to foster a more relaxed, conversational environment. Additionally, these self-disclosures were used strategically to direct the conversation, encouraging further discussion or elaboration on certain topics.

Interviews began with an introduction script outlining the researcher's role and key study details. After explaining the study's purpose, confidentiality measures, and participants' rights, verbal consent was obtained for participation and audio recording. Demographic questions were asked, with those on gender identity and sexual orientation positioned at the end of the section to recognize the sensitivity of these topics in Belize's conservative Christian context, where many people conceal such identities due to social stigma. Building rapport before discussing these topics was essential, as many queer Belizeans are only "out" to trusted friends.

The core of the interview focused on interviewees' experiences of adversity. Follow-up questions explored coping strategies, available support, personal development, and subsequent personal changes. Probes and prompts were used to encourage deeper reflection and detailed responses. Closing questions captured any additional thoughts, and a closing script outlined the next steps and reiterated confidentiality. Finally, participants received a follow-up email thanking them for their contributions and providing a debriefing form (See: Appendix C).

Reflection on Interviewing Close Friends, Acquaintances, and Strangers

Interviews with friends differed from those with acquaintances or strangers in several ways. While rapport was already established with friends, the interview content –

particularly discussions of past adversities – was unfamiliar within the context of those relationships. The level of caution in asking follow-up questions varied depending on the friend and the topic, as the interviewer aimed to avoid offending or coercing responses. In contrast, caution remained consistently high when interviewing strangers or acquaintances. Friends also tended to offer more targeted responses they believed would be helpful. Interviews with acquaintances and strangers felt different in nature, with fewer concerns about straining a relationship shaped by pre-existing boundaries or patterns of communication.

Throughout the data collection period, the researcher engaged in regular meetings with their primary supervisor, a clinical psychologist with a strong background in qualitative research and clinical interviews, as well as the assessment and treatment of trauma. These meetings allowed the researcher to discuss progress, troubleshoot potential issues, and receive guidance and feedback.

Data Analysis Procedure

An Interpretative Phenomenological Analysis (IPA) approach, as outlined by Braun and Clarke (2021), was used to understand the participants' lived experiences and their subjective interpretations. The analysis procedure was carried out manually, using a combination of pen, paper, and a word processor. It consisted of two phases: (i) individual analysis of transcripts and (ii) cross-case analysis.

First, the researcher read a transcript multiple times to become familiar with the content. Notes were taken on the author's initial impressions, thoughts, and reflections about the participant's experiences and expressions. Significant statements, phrases, or sentences relating to the research questions were then highlighted. Following this, the contents of the notes were grouped into emergent themes. To do this, the author interpreted how different notes relate to each other and organized them into coherent clusters that reflected significant aspects of the participant's experience and how they interpret it, as well as how it contributes to the participant's overall narrative. This procedure was utilized with each transcript (N = 12) individually and sequentially.

After this analytic process, a cross-case analysis was performed to identify common themes and variations across participants. Similarities and differences across

participants' experiences were systematically examined, yielding a list of super-ordinate themes. Finally, drawing on the principles of grounded theory, an inductive approach was used to expand and refine Krupnik's Hybrid Model of Stress Response (HMSR).

Findings

The participants' gender identities were diverse, with ~41% identifying as female, ~33% as male, and 25% as non-binary or gender-fluid. Their ages ranged from 22 to 44 years ($M = 29.7$, $SD = 6.5$). In terms of sexual orientation, ~41% identified as straight, ~41% as bisexual, pansexual, or polyamorous, and the remainder as gay or on the asexual/aromantic spectrum. While this sample reflects a broad range of identities, it is not representative of the general population. Due to the socio-political and religious climate, the true proportions of gender and sexual diversity in the general Belizean population remain unknown. However, regional comparisons may offer some insight. For instance, in Mexico, a neighboring country with a predominantly Catholic population (Rainsford, 2023), a national survey estimated that 5.1% of people identify as LGBTI+ (INEGI, 2021).

Ethnically and culturally, a majority identified as Creole, some as Mestizo, and a few as Mennonite, Guatemalan, Indian, Nigerian, or a mix of these backgrounds. All participants speak Kriol and English, 50% also speak Spanish, and a few speak additional languages such as Low German, German, or Igbo. Geographically, most are based in/by Belize City or Belmopan, with a couple in regions such as Orange Walk (Belize) and North America. Two participants living abroad have been away for 0.5 to 6 years, but continue to visit regularly.

Beliefs and practices-wise, a few participants identified as religious Christians, while the majority identified as spiritual non-religious Catholic, custom spiritual, agnostic, or atheist. In the latter group, almost all of these individuals had previously been involved in Christianity but moved away from the faith due to personal reasons, often linked to experiences of adversity, disillusionment, and/or marginalization with the church and associated religious circles. Some of these individuals now follow a blend of indigenous and alternative spiritual beliefs/practices, which hold deeper personal meaning, empowerment, resilience in the face of adversity.

Overall, participants reported a wide range of adversities, including studying abroad, mental health struggles, and perpetrator-based violence. Approximately one-third of the participants reported experiencing sexual assault (SA), all of which occurred during childhood or adolescence. One individual reported being subjected to both SA

and death threats. Additionally, a majority of participants reported experiencing emotional and/or physical abuse or harm from interpersonal relationships. Due to time constraints, all participants chose to focus on one to three major adversities in their lives. Some mentioned that there were many other significant experiences they chose not to discuss, as time was limited and they prioritized sharing the most impactful ones.

Three Major Themes Identified Through IPA

The following major themes and sub-themes were identified from the interviews: (i) *long-term self-regulation disruption*: psychopathology, cumulative stress from emotional avoidance, helplessness and powerlessness; (ii) *adversity intensified by perceived failure of support*: absence of support, denial of support and rationalizing harm, harm instead of safety; (iii) *coping and social support mitigating the severity of adversity*. Notably, for all but one participant, these themes arose organically in participants' narratives before the interview question "What sort of support was available?" was asked. The majority of participants described concurrent patterns related to Theme 2, sub-themes 2 and 3, as well as that of Theme 3. In response to the question "How did you deal with this experience?," when participants did not report using support-seeking as a coping strategy, they described manifestations of psychopathology, such as depression, insomnia, and pervasive negative affect.

Theme 1: Long-Term Self-Regulation Disruption

Among the participants, the long-term psychological effects of trauma and adversity are most evident in the disruption of self-regulation. The emotional and psychological manifestations of stress responses – whether panic attacks, low self-esteem, lack of agency or control, powerlessness, or helplessness – often creates an ongoing source of distress that impacts participants' ability to function in their daily lives. The disruption of self-regulation experienced by participants was organized into three key sub-themes: psychopathology, cumulative stress from experiential avoidance, and feelings of helplessness and powerlessness.

Sub-theme 1: Psychopathology

In this context, psychopathology refers to behaviors or thought processes that significantly impair an individual's ability to function (American Psychiatric Association

[APA], 2022). These impairments involve disturbances in cognition, emotional regulation, or behavior, causing substantial distress or disability in key areas like social, occupational, or daily activities. Culturally expected responses to common stressors or losses are excluded from this definition.

In the present study, the instances of psychopathology described by participants were consistently linked to the stressful events they recounted. Notably, only one participant reported a trauma-related diagnosis – complex PTSD (cPTSD). About half of the total sample described severe and chronic episodes of anxiety and/or depression impairing daily functioning, although only a minority reported formal diagnoses. In this group, for individuals without a formal trauma diagnosis, the post-stressor responses reported were pervasive and distressing, consisting of coping mechanisms such as daydreaming, emotional numbing, and an aversion to intimate or close relationships. Several participants reported pervasive feelings of distrust, guilt, and self-blame, leading to isolation and emotional withdrawal. Persistent negative emotional states and the inability to experience positive emotions were commonly described, along with episodes of memory loss related to the period of time surrounding the stressful event. Some participants also continue to experience hypervigilance and sensitivity to stimuli, such as loud noises or physical contact, and sleep disturbances, including insomnia and restless sleep.

Notably, a few participants did not report experiences or diagnoses of psychopathology, even though all participants reported multiple highly distressing experiences. It is important to note that the interviewer avoided probing deeply into participants' subjective experiences of the adversities themselves, due to potential discomfort for participants. As a result, all descriptions of psychopathological experiences are based solely on what participants were willing to report during their interviews. If the interviewer asked about psychopathology, it was solely to clarify or better understand what participants had already mentioned. These questions were provided using the language that participants themselves had used. Thus, these findings reflect only what the participants felt comfortable disclosing, rather than a comprehensive or detailed account of their experiences with those adversities or of their broader life experiences.

Sub-theme 2: Cumulative stress from experiential avoidance

Many participants reported that experiential avoidance – the avoidance of internal experiences such as negative emotions relating to the stressful event – contributed

to cumulative stress over time, which increased vulnerability to subsequent stressors. This accumulation of unaddressed emotional pain, compounded by later events, can overwhelm self-regulation capabilities, often leading to emotional and psychological breakdowns. As one participant, Sasha, stated:

[I]t was many years that I hadn't dealt with a lot of my own inner emotions, like, throughout my teen years and whatever. And sometimes, sometimes you're just trying to survive. So like, sometimes you are literally in survival mode for a long time. And like, I think after many years of survival mode, my body was just like, I can't do this anymore. (Sasha)

Throughout her life, Sasha faced severe adversities, including parental divorce and sexual abuse in early childhood. Her avoidance of emotions related to these experiences eventually took a toll on her well-being, resulting in severe depression in young adulthood. Many studies have suggested that emotional avoidance may develop and maintain trauma-related symptoms (see: Meuleman et al., 2024; Pineles et al., 2011; Schick et al., 2020).

This sub-theme expands upon the HMSR framework by showing that self-regulation breakdowns are not only the result of extreme stressors themselves, but can also stem from individuals' responses to these stressors, particularly through experiential avoidance. Furthermore, participants' experiences suggest a temporal dynamic in the nature of self-regulation, where the impact of avoidance is not static. Over time, continued avoidance can exacerbate stress, leading to a cyclical pattern where emotional and psychological breakdowns become more frequent and difficult to manage.

For instance, Ruth describes the following:

[E]verything [was] fine... [but] why do I keep having panic attacks... I would be in class and then all of a sudden, and crying and having a huge panic anxiety attack... it was starting to happen at work. It happened when...really embarrassing. I was like, talking to this guy...like a cloud just came over me. (Ruth)

Despite the traumatic experiences, as Ruth labels them, occurring in her childhood and adolescence, the dysfunction of self-regulation worsened significantly during early adulthood, manifesting in severe panic attacks. This deterioration occurred during a period in which Ruth reported considerable experiential avoidance relating to her earlier traumatic experiences. This suggests that trauma responses can increase in severity as time

progresses, maintained and/or exacerbated by experiential avoidance, even years after the initial traumatic events.

Experiential avoidance and its connection to trauma are well-established in both research literature and clinical practice. For example, the DSM-5-TR identifies avoidance of internal experiences (e.g., memories or emotions) as a core symptom of PTSD (APA, 2022). While the directionality of the relationship between emotion regulation difficulties and PTSD remains unclear, as most studies have been cross-sectional, both possibilities are supported by research (Meuleman et al., 2024). On one hand, PTSD symptoms may drive individuals to adopt maladaptive strategies, like avoidance, creating a cycle that perpetuates distress. On the other hand, emotion dysregulation can increase the perception of threat, making stressors feel overwhelming and contributing to PTSD.

Some participants have also experienced the latter case, where emotion dysregulation has made stressful situations more overwhelming. One participant, Maestro, for instance, described an experience performing music abroad at a young age:

I felt like I didn't belong, which made me feel very inadequate...[they're] emotionally probably more mature than me... we were in a group [walking]... and I just remembered going off on my own...I don't think the group is gonna wait for me or ... wants me around. So I'm just going to find my way back on my own. And that was bad because they got frightened...they couldn't find me. They were like, "Where the fuck did you go?" (Maestro)

While she did not label this experience as trauma, she described it as a very psychologically stressful event. The emotional stress that Maestro faced during this period is consistent with the idea that emotion dysregulation can make stressors feel more threatening, thereby exacerbating their impact. Moreover, this dysregulation could lead to experiential avoidance behaviors, such as withdrawal or self-isolation, that could also intensify the severity of the stressor itself. In Maestro's case, the situation became even more stressful when the group reacted negatively to her withdrawal, which could have reinforced feelings of inadequacy and potentially led to self-blame or guilt.

Sub-theme 3: Helplessness and powerlessness

For many participants, panic attacks and severe depressive episodes continued to affect their daily lives long after the adverse event, often rooted in feelings of helplessness and powerlessness. These persistent emotions and symptoms were linked to past

adverse events. For instance, in relation to the increasing frequency of panic attacks in her young adulthood, Ruth said, “it would feel like I didn’t have any control.”

Similarly, Anna has experienced unexpected panic attacks well into adulthood:

I don’t know what to do... I can't even think for myself anymore [during abusive relationship (AR)]... [after AR] I thought I wasn't even a proper adult, like I didn't know how to do a lot of things. Just going to the bank gave me a panic attack because I didn't know how that worked. (Anna)

This ongoing struggle suggests that unresolved psychological processes in relation to severe adversity, whether emotional, cognitive, or physiological, remain a significant barrier to individuals regaining a sense of agency and stability in their lives. These narratives show that these unresolved feelings often generate new sources of stress or contribute to additional adverse experiences. In other words, consistent with existing literature, these commentaries suggest that trauma exposure can increase the risk of exposure to more highly stressful or traumatic events later on (Solomon et al., 2021).

Theme 2: Adversity Intensified by Perceived Failure of Support

One of the most profound aspects of participants’ experiences was the betrayal of trust in individuals or institutions that were expected to provide safety and support. Whether in familial, institutional, or community contexts, the absence, withdrawal, or denial of support can intensify feelings of distress.

Sub-theme 1: Absence of support

For example, Miles, a high school teacher in South-Side Belize City, spoke of the immense stress of losing students to gun violence, with the deaths occurring outside of school, compounded by the lack of institutional mental health support for staff:

[T]wo of our, of our students were killed...the year after next, one was killed, and then this year one had passed away ... child was, was in, was in the class I was teaching ... I had to console my [students]... would have been nice if a person was there ... for us staff... it was a very hard thing to handle... And so it kind of felt bad that, I mean, they [the students] had to open up to me ... I'm not really a person who is trained in that, right? (Miles)

In addition to supporting his students through grief, Miles also found himself providing support to students who confided in him about past experiences of sexual assault, many

occurring as young as eight years old and younger. Teaching, overall, is a significant and chronic source of stress – so much so that he wishes to change professions. Miles has had to rely on a friend, a counselor, for emotional support (not formal counseling). However, due to the national shortage of mental health professionals, many teachers, unlike Miles, would not be fortunate enough to have friends who are mental health professionals, leaving them without vital support.

Sub-theme 2: Trusted sources denying support and rationalizing harm

In other cases, participants reported the denial of support and victim-blaming from trusted figures, such as family and religious institutions. For instance, Sage shared their experience of being told by their father that sexual assault was a normal part of life, minimizing the significance of the perpetrator's actions:

I had told my mother about it. But only because she caught me crying ... my dad did try to have a conversation with me about it afterwards... it went along the lines of like, you know, men just can't help it... that was like the breaking point for me. Just realizing that I did not have any support. (Sage)

To compound a lack of parental support, participants' church and community (inter-changeable terms in this context) pushed the narrative that the responsibility for the harm laid with them:

I went to like leaders in the church...Their solution was simply to have counseling sessions with the man and his family. And that was it. There were no repercussions for him... and they have this whole discourse of like, what did you do to tempt him? And that that was the first time that that happened, that when I was 15... the church is supposed to be there for each other, right? Like the members. But in the end, there was no one there for me (Sage)

In Anna's experience, despite her visible injuries and emotional distress, her mother supported the continuation of her abusive relationship in favour of potential financial stability:

I was in a very bad abusive relationship... ended up telling my mother ... she didn't do anything ... growing up in a family that didn't have much, meeting someone... that had money and had a good living, her opinion was like, "Oh, you have to do anything you can to stay with this person, because I don't want you living the life that I did"... But like this person is like physically hurting me, and this is what you're telling me... the main person that I thought should have told me, "okay, well, we can do something about it, or I'll help you." She didn't. (Anna)

Despite the “normalization” of harmful behaviour by trusted sources, such as parents and church clergy or members, the negative impact of these individuals’ adverse experiences did not lessen. According to Sage, “what exacerbates the stress of the trauma is not being able to have a safe place to go and thinking that you do.” These narratives suggest that trauma may not only be exacerbated by the refusal of support from close family and social networks but also by the invalidation of the adverse experience. This failure closely aligns with existing research, which shows that general and trauma-specific invalidation predict the development of psychopathology, such as anxiety, depression, PTSD (Hong & Lishner, 2016).

Sub-theme 3: Harm instead of safety

This sub-theme illustrates the deeply troubling instances where individuals, particularly those who were supposed to offer care, protection, and safety, instead contributed to or directly caused harm. For example, in Victor’s experience, despite being raised in a very sheltered and supposedly safe environment, around only family members:

[W]hen I was younger, being like, introduced to, like, sexuality by one of my older cousins...I was very sheltered and protected as a kid, and not really allowed to, like go out and like, be at my friend's houses... I was like, you know, trying to find friends outside of just like, my cousins being like, my friends the only friends. (Victor)

Victor’s reflection shows a dissonance between the expectation of protection (sheltered and “protected” childhood) and the sexual violation of that safety by the very person who should have been a trusted source of guidance and care.

Ruth also described a similar experience, where the people responsible for her care betrayed her trust in a traumatic way. The situation, which was supposed to be a safe environment, instead became a site of exploitation and harm, where her babysitter gave permission for the adverse events to occur:

It was actually my babysitter and her boyfriend. Um, and you know, they're supposed to be people as taking care of you but you know... I'm I'm so disoriented. I get woken up... they're having a conversation about me... and then she was just like, well, this is my job, and don't get in trouble. And then and then I felt pain... (Ruth)

This suggests that when safety and care are replaced by harm, the distress is not only a result of the traumatic event, such as sexual assault, but also of the profound breach of trust. Additionally, when the perpetrators of the trauma are in the positions of caretakers, this may lead to severe sustained stress due to feelings of betrayal, helplessness and powerlessness, self-blame, shame, guilt, and fear of repeated future incidents.

Theme 3: Coping and Social Support Mitigating Severity of Adversity

While stress responses can feel overwhelming, participants also highlighted the significant role that coping mechanisms and available support systems played in mitigating the severity of their experiences. Support, both internal (such as self-reflection and journaling) and external (such as peer and familial support), emerged as key factors that lessened the long-term psychological damage of traumatic events. For instance, Ruth described how journaling helped in self-expression and gaining self-agency and how family and peer support made things easier to handle.

...journaling really helped me process my emotions, and kind of still be able to have a voice and speak as loud as I wanted to without disrupting people. So journaling helps me a lot, you know, talking with my brothers and sisters, I don't know if it was trauma, bonding, or whatever, but we would be like, together, experience we had each other, you know, so we can make jokes about it or whatever, and it would be okay. (Ruth)

Moreover, Anna spoke about the significance of peer support from friends, which played a crucial role in her ability to leave an abusive relationship:

So because of them [friends], I ended up coming out of that relationship. And because if it was left up to me, just me, I probably think I wouldn't be alive today. (Anna)

These accounts illustrate the importance of supportive networks – not only for emotional relief but also for survival and recovery through traumatic events.

Connecting Theme 2 & 3 to Literature

For the same individual facing the same adverse event, would the lack of expected social support make the experience feel traumatic or more traumatic, while strong support could make it highly stressful but not necessarily traumatic? According to a meta-analysis of longitudinal studies, social support and PTSD are closely

interconnected in a reciprocal way: social support reduces PTSD symptoms over time, while PTSD also predicts a reduction in social support (Wang et al., 2021).

This reciprocal relationship suggests that the presence or absence of support may influence how severely an adverse event is perceived and processed. Let us assume that PTSD symptoms are seen as markers of trauma severity, with a greater number and/or heightened intensity of symptoms indicating a more severe traumatic stress response and thus, a perception of the adverse event as more severely and negatively impactful. In this light, the longitudinal findings would suggest that strong social support may act as a buffer, reducing the likelihood that an adverse event will be perceived as deeply traumatic and instead framing it as a stressful but manageable experience.

Wang et al.'s (2021) findings strongly support three major socially-oriented stress-distress perspectives: (i) the social causation theory, (ii) the stress-buffering model, and (iii) the social support resource theory. According to the social causation theory, social resources, such as support networks, are essential predictors of well-being, particularly in the context of stress. Trauma, in this view, disrupts the individual's social sense of self and connection to others or the world, while recovery or resilience involves the restoration or strengthening of these social bonds.

The social support resource theory complements this view by framing social support as a resource that reinforces self-identity and provides additional reserves to buffer against stress. Access to social support enhances an individual's capacity to cope by substituting for other depleted resources. Similarly, the stress-buffering model posits that social support acts as a protective factor, helping individuals positively or constructively reappraise and manage the demands of stressful or traumatic experiences.

Thus, the findings in Theme 2 and 3 above closely align with these perspectives. The perceived absence of social support might increase the risk of more severe PTSD symptoms, intensifying the perception of the event as traumatic; conversely, the presence of support reduces the perceived severity of the demands or impacts of the stressful event.

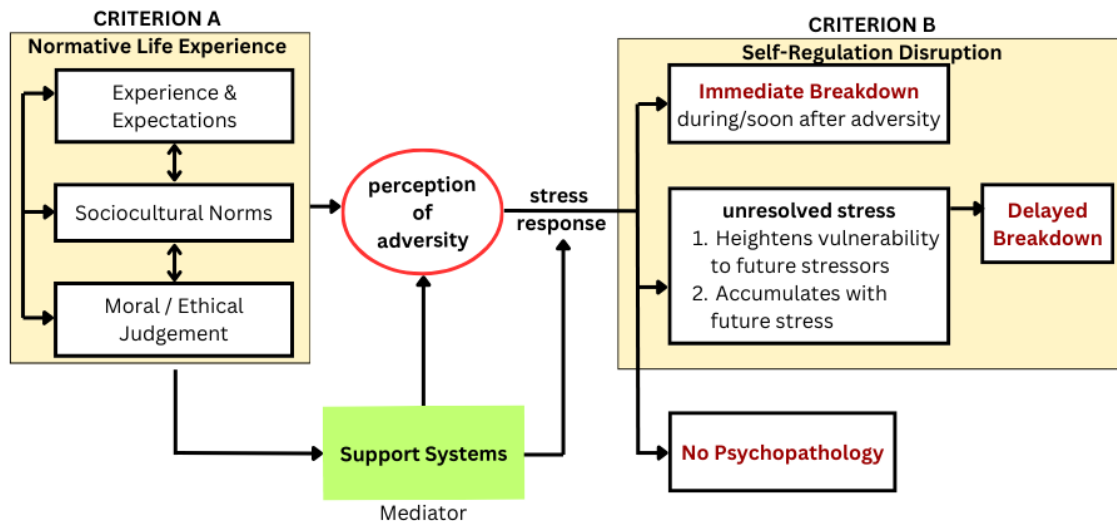
Wang et al.'s (2021) findings also offer insights into Theme 2: Sub-theme 2, by demonstrating that psychological distress can negatively affect the availability or quality of social support resources. Several prominent factors contribute to this, including the

interpersonal difficulties, distorted perceptions of available support, and self-isolating behaviors often associated with PTSD. Thus, taking these insights and the three social-based stress-distress perspectives into account, the absence of support from support networks can greatly exacerbate the psychological consequences of trauma. Moreover, the negative effects of being refused support may be more severe than the mere absence of support, such as seen in Anna’s experience.

An Expanded Hybrid Model of Stress Response

Building on these findings, I propose an *Expanded Hybrid Model of Stress Response* (E-HMSR; see Figure 4) within a process framework that aims to better define “trauma”. The E-HMSR retains and expands upon Krupnik’s foundational criteria of *Normal Life Experience* and *Self-Regulation Disruption*, as well as the addition of *Support Systems* as a critical mediating factor that shapes both criteria and influences individual outcomes. These proposed components assist in redefining Krupnik’s boundary between pathogenic and traumatic stress responses.

Figure 4. A Process Framework Illustration of E- HMSR



Criterion A: Normative Life Experiences

As participants' narratives illustrate, the concept of what constitutes "normal" varies significantly across cultural and individual contexts. Many Belizeans described adversities – such as sexual assault, "whipping" as a childhood punishment, or systemic oppression – as "common" and even "expected," reflecting a normalization of adversity within their communities. However, despite the normalization of these stressors, the impacts are still found to be immense and negative, as described by Anna:

...now... talking to like other people, they're like your your mom wasn't supposed to do that. That's considered as abnormal, whereby we found it as a very normal thing, but yeah... that was like, I think the biggest thing that affected me psychologically, now today, as a person. (Anna, on her mother encouraging her to stay in an abusive relationship)

This account and formulation challenge Krupnik's assumption that trauma stems from non-normal experiences. This same pattern can also be seen in Sage's experience, where the perpetrator's actions were normalized and justified according to their religious community's beliefs. In Mile's case, it seems that student deaths from gun-based violence and emotionally supporting students through very challenging events (e.g., friends dying, childhood SA) have become a "normal" part of his profession, yet it has continued to be immensely stressful.

It is evident that the "normal life experiences" criterion in Krupnik's model requires refinement and expansion. To address this, I propose relabeling the criterion as *normative life experience* to emphasize the subjective perception of the individual. This shift moves away from defining normalcy based solely on statistical or societal norms, focusing instead on how the individual interprets the stressful event in relation to what they perceive as typical or expected. To expand upon this criterion, I propose integrating three dimensions: experience and expectations, sociocultural norms, and moral/ethical judgment.

Experience and Expectations

Personal expectations and prior experiences can shape how individuals perceive and respond to a stressful event. What may seem manageable to one person could feel overwhelming to another, depending on their history. For instance, despite the intense

pain of kidney stones, Ellie viewed the event as something she could withstand, drawing on her past experiences.

[T]he Easter after [giving birth], I passed a kidney stone on my own at home, a little bigger than the size of an olive pit.... And it was painful, yes, but I could withstand it ... didn't feel like [description of labour pain] And so I think that's like a physical equivalent of the emotion that I felt in that past. And so I think knowing like whoa, things could hurt... so much [more]. (Ellie)

Thus, individuals who have experienced significant adversity may view subsequent challenging situations as more manageable, while those with less exposure may find the same events traumatic.

Sociocultural Norms vs Moral Judgment

Even though certain events or behaviors might be normalized or even expected within a particular society, they may still feel stressful or traumatic to the individual based on their personal context and values. For instance, growing up in a church community where the nuclear family was the ideal, Sasha struggled with the experience of having an absentee father:

I think it matured me very early, because I had to learn a lot of things that maybe like other kids don't have to worry about at that age... having to deal with an absentee father, like, you know, many times kids have, you know, a like, a whole family, um, and really, what is a whole family? Because he could have died, so, you know. (Sasha)

Yet, the prevalence of family instability in Belize reveals the subjective and context-specific nature of this experience. While Sasha's family situation deviated from the norm within her church community, the broader national social context shows that family instability is widespread.

Additionally, even if an event is normalized within a society, it can still be perceived as traumatic if it violates an individual's moral or ethical framework, especially when interpersonal violence and/or harm are involved. For example, in Sage's case, despite efforts by their community to normalize and downplay the impact of their sexual assault, the event remained inexcusable and profoundly traumatizing.

For some individuals, even if they did not initially see a stressful event as a violation of their morals at the time, it can still be recognized as harmful through the awareness of negative psychological consequences. For instance, in Frantz's experience:

[A]nything we did, we'd be whipped... everyone ...that's the way they raise their kids...It just creates a fear and anxiety that when you do something wrong...you'll be violently held, held to account, but you won't actually understand what you did wrong ... so I identified with the word whip ... is carrying on of just whipping black bodies... the underlying notion that guided all of this is ... they're doing it out of love...this is the support...the argument they would, they will make and that's also the framework I would have had back then... I don't agree with it at all [now] ... it created, uh, a barrier ... I'm not fully able to call to talk to my parents... when I do need anything, there is this... anxiety. (Frantz)

During his childhood, Frantz accepted the belief, shared by his parents, that whipping was a form of support and love, as well as “normal” discipline. Despite this, it has negatively affected his relationship with his parents, as well as contributed to stress responses that have persisted into his adulthood, such as anxiety and avoidance.

Criterion B: Self-Regulation Disruption

In this expanded model, I propose that self-regulation capabilities should be understood as a dynamic process that exists along a temporal continuum, rather than as a static trait. The breakdown of self-regulation is not a singular event but evolves over time, shaped by both internal and external factors. This model distinguishes between two key types of self-regulation breakdown: immediate and delayed.

The immediate breakdown refers to the acute disruption of self-regulation that occurs during or shortly after a traumatic event. In contrast, the delayed breakdown happens when the effects of trauma do not initially overwhelm the individual's self-regulation abilities, but instead gradually erode them over time. These categories are not mutually exclusive; an individual may experience an immediate breakdown following a traumatic event, which may later be followed by a delayed breakdown as unresolved stress accumulates. This distinction aligns with broader trauma research showing that responses to adversity, specifically PTSD, fluctuate over time rather than following a fixed pattern. Some individuals recover quickly after initial distress, while others experience a delayed decline in functioning or chronic distress from the outset (Bryant, 2019).

Mediating Factor: Support Systems

The findings suggest that support systems play a crucial mediating role in determining the psychological outcomes of adversity. While trauma and stress can have profound and lasting effects on mental health, the presence or absence of support from key social networks – such as family, friends, religious communities, or institutions – can significantly influence the severity and long-term impact of stress responses. This observation suggests that trauma should not only be understood in terms of the normativeness or post-regulatory effects of the event itself, but also through the lens of social support structures.

Interaction with Normative Life Experiences

Support systems can interact with the individual's perception of what is "normal" in relation to stressful events. When support is refused or unavailable from key social networks, the experience of stress can become even more traumatic due to a perceived moral or social betrayal. This outcome may stem from the individual's belief that their suffering is not recognized or validated by those they rely upon, leading to a sense of rejection or alienation and enstrangement. On the other hand, support that aligns with the individual's moral framework can have a protective and positive effect.

Additionally, individuals may experience greater recovery and resilience when buttressed by support systems that challenge socio-cultural norms. In the case of one participant, the support from his aunt became a pivotal factor in his recovery and resilience, after being "outed" to his family and abandoned by his partner. His mother not only rejected him but also actively prevented his extended family from offering support, leaving him homeless and without any security or safety net.

I called my aunt and I told her listen, my mom kicked me out...It took me a while before I can say Tia I'm gay ... she's like, what's wrong with that? Nothing's wrong with that... Just come home. Don't worry about anything... that made a difference...I can now feel safe that I will have somewhere to go and not worry about nowhere to go ...and no one to help me out... my guardian angel at that point was my Aunt. (Richard)

In a situation where all expected support systems had rejected him, the acceptance from his aunt became the source of his resilience and recovery. Notably, this participant is one of the minority in the study who did not describe experiences of psychopathology despite enduring significant adversity.

Interaction with Self-Regulation

As seen in Richard's case, as described above, the quality and presence of support systems can also significantly affect self-regulation in the face of severe stress. Strong and consistent support can act as a buffer, mitigating the erosion of self-regulation and preventing the buildup of unmanageable stress. For instance, a strong support system can provide emotional validation and a safe space to process the adverse event, despite ongoing stressors. Conversely, as seen in Theme 2, the lack of support exacerbates the stress experienced during the adverse event, negatively impacting self-regulation capabilities.

Pathogenic vs Traumatic Stress Responses

In the Expanded Hybrid Model of Stress Response (E-HMSR), pathogenic stress responses (PSR) and traumatic stress responses (TSR) are conceptualized as points along a dynamic continuum rather than distinct and exclusive categories. Participant narratives highlight that stress responses can evolve over time, shifting from pathogenic to traumatic responses as internal and external factors interact with the individual's ability to regulate stress. For instance, as seen in some participants' narratives, prolonged exposure to stressors or unresolved ongoing stress can escalate a PSR into a TSR by gradually eroding the individual's self-regulation over time.

PSR refers to stress that leads to pathology, such as anxiety or hyperarousal, but does not immediately disrupt self-regulation. In contrast, TSR involves significant breakdowns in self-regulation, resulting in severe psychological dysfunction. However, the boundary between PSR and TSR is not always clear, as symptoms like emotional numbing and hyperarousal may appear in both. This overlap raises questions about the nature of self-regulation in PSR versus TSR. Is self-regulation truly intact if anxiety is present? Perhaps the distinction lies in the severity and manageability of symptoms – while anxiety within PSR may be manageable, in TSR, it can be debilitating, overwhelming coping mechanisms entirely. Further exploration is needed to refine this distinction and better understand the thresholds that separate these responses.

Practical Implications of E-HMSR in Belize

Given the limited sample size of the present study, further data collection is crucial before practical implications of the proposed E-HSR can be confidently suggested. However, preliminary insights suggest the need to reassess mental health screenings in Belize to ensure that they account for the role of support systems in individuals' well-being. Current evaluations may overlook the impact of both formal and informal support networks, which emerged as significant influences in the participants' experiences. Furthermore, community-wide support structures should be systematically assessed to identify gaps and inconsistencies, as some participants reported drastically contrasting experiences regarding church communities.

Assessing these points in mental health care can inform the design of targeted interventions. For instance, insights from these assessments could inform the development of community-based religious support programs that emphasize positive religious and spiritual coping strategies. Such programs could be tailored to benefit children and parents, fostering resilience and emotional well-being through faith-driven practices that align with Belize's cultural context.

Strengths and Limitations

While the findings provide a valuable starting point, further research is necessary to ensure a more comprehensive understanding of the range of adversity and trauma experienced within Belize. The findings are not generalizable to the broader population due to several methodological limitations. Although the sample size of 12 participants is appropriate for an Interpretative Phenomenological Analysis (IPA) study, it cannot capture the full range of experiences across the country. The requirement for English fluency and access to Zoom for the interviews potentially limited participation from certain groups, particularly those who speak languages other than English or those without reliable internet access, which could skew the findings toward more urban, educated, and connected individuals.

Additionally, the sample was notably diverse in terms of gender identity and sexual orientation, providing insight into a range of lived experiences often underrepresented in psychological research, especially considering the broader context of social stigma and invisibility faced by queer individuals. Although the sample cannot be considered representative, the inclusion of these voices is a strength of the study, as it may highlight forms of adversity that may otherwise remain overlooked.

Several potential biases should also be considered. The use of snowball sampling, while effective in identifying participants who may have otherwise been difficult to reach, could have introduced biases in participant selection. The participants' networks may have influenced who chose to participate, and individuals who are less socially connected or less comfortable discussing personal adversity may have been excluded. On the other hand, this recruitment strategy may have also helped include these individuals, as they may have chosen to participate due to trusted connections within their networks.

Self-selection bias is another concern, as participants willing to discuss their personal experiences may differ from those who chose not to participate, potentially resulting in an overrepresentation of individuals who are comfortable discussing their adversities. Nonetheless, purposive sampling helped ensure diversity in terms of age, gender identity, and sexual orientation, and allowed for a range of perspectives to be included in the study. Despite the emotionally-loaded and often traumatic content shared during the interviews, all participants reported a sense of relief, catharsis, or appreciation for the

opportunity to share their experiences. This outcome suggests that, while the content may have been distressing, participants directly benefited from participating, highlighting the potential benefits of talking therapy for many Belizeans.

Additionally, the study utilized secondary data analysis, which offered several advantages, such as minimizing the need for new data collection and preventing participants from revisiting sensitive topics. However, this method also posed limitations. Secondary data analysis meant that the research questions and interview protocols were constrained by the design of the original study, which focused on adversity, coping, available supports, and post-traumatic changes (e.g. growth). This factor limited the scope for exploring deeper questions related to trauma, adversity, and mental health that were central to the present study. As a result, participants may have been more inclined to focus on positive aspects of their experiences, such as post-traumatic growth, while possibly downplaying struggles or negative emotional experiences.

This orientation could have led to a potential under-reporting of psychopathology, emotional distress, and/or challenges in coping – all relevant areas when examining and expanding upon Krupnik's model. While the data limitations may have restricted the ability to fully expand on Krupnik's model and could have affected the validity or applicability of the findings, the proposed model still provides invaluable insights into stress and traumatic responses, providing a foundation for further exploration and refinement of the model in future research.

Future Research Directives

A particularly thought-provoking avenue for future research arises from a participant's offhand remark at the conclusion of their interview. The participant described witnessing a gang-related gunfight and seeing a dead body near their residence, in their adolescence. Remarkably, they reported no emotional reaction to the event – either at the time of the incident or in retrospect. This experience raises questions regarding how youth, especially those frequently exposed to violence, process and respond to potentially traumatic events.

One possible explanation is desensitization, as research suggests that medium to high levels of real-life violence exposure may reduce empathy and emotional responses to future violent events (Mrug et al., 2015). In environments where violence is pervasive, individuals may adapt by emotionally disengaging as a coping mechanism, leading to a muted or absent reaction to events that might otherwise be perceived as distressing. In the present study's interview protocol, adversity was framed as “very psychologically stressful events.” Thus, it is possible that some participants, despite having been personally exposed to physical harm and death, did not perceive these events as sufficiently distressing to discuss.

To explore how Belizeans' perceptions of trauma are impacted by desensitization, a cross-sectional study could be carried out using a combination of the Trauma History Questionnaire (Hooper et al., 2011), an adapted version of Galán Jiménez et al.'s (2019) Desensitization to Violence Scale for adults, and the PTSD Checklist for DSM-5 (PCL-5; Blevins et al., 2015). The primary aim of this study would be to explore how trauma history is related to PTSD symptoms, and to examine whether desensitization to violence moderates this relationship. Specifically, the study would investigate whether individuals who report higher levels of desensitization show a weaker correlation between trauma exposure and PTSD symptoms compared to individuals with lower levels of desensitization.

Qualitatively, to explore whether certain potentially traumatic events are perceived as distressing in Belizean contexts, cognitive interviews could be conducted to explore participants' interpretations of trauma-related questions. These interviews would use structured questions based upon the Trauma History Questionnaire, through

contextual probes (“Can you describe a time when you saw [event] happen but didn’t consider it stressful?”) and cultural probes (“Do you think most people in your community would see this event as traumatic? Why or why not?”). These interviews would help identify cultural and contextual influences on trauma perception.

Through these described methodologies, future research could also explore which Belizean demographics are the most or least vulnerable to trauma exposure and trauma-related psychopathology. Perhaps emotional desensitization may play a role in greater resilience – though research has shown that when this occurs in children, it is associated with greater impulsive and aggressive behaviour (Pat-Horenczyk et al., 2014). Identifying at-risk populations and protective factors can inform the development of targeted public health strategies, educational programs, and governmental policies to mitigate the negative psychological consequences of adversity.

Overall, these proposed research directives would also directly contribute to and expand upon the E-HMSR proposed in the present study. These quantitative and qualitative approaches could reveal how desensitization impacts trauma perception and stress responses, which may have important implications for understanding how individuals in Belize, as well as similarly high adversity contexts, process and respond to potentially traumatic events.

Conclusion

Overall, it is concerning that all participants in this study reported experiencing multiple extremely stressful events, with one-half describing severe and long-term post-stressor symptoms, manifesting in the form of self-regulation disrupting symptoms. This raises significant concerns about the mental health needs of the Belizean population, given Belize's limited mental health infrastructure and support systems.

Despite some limitations, the present study has contributed significantly to the understanding of adversity and trauma in Belizean adults. Given the complexity and context-sensitivity of trauma, it is clear that the criteria involved in defining trauma should be equally complex and context-sensitive. As shown in the findings of this study, adversities that may seem "normal" or "expected" in certain cultural or community contexts can still have profound traumatic effects. By broadening the scope of trauma criteria to account for these contextual elements, we can develop a more accurate and inclusive framework for understanding the diverse ways in which individuals experience, process, and cope with trauma. Such an approach would not only deepen psychological understanding but also create more culturally relevant interventions and support structures for those affected by trauma.

Participant narratives have revealed how the absence of support or its limitations, or the presence of harmful support systems, can exacerbate the impact of adversity, suggesting that trauma responses are not only influenced by the nature of the event but also by the surrounding social environment. The primary goal of this study was to lay the groundwork for a more culturally-informed understanding of trauma, providing a foundation for future research to explore how trauma can be better defined and addressed within the Belizean context. In doing so, it not only contributes to the growing body of knowledge on trauma in culturally diverse and historically underrepresented populations, but also offers important implications for creating culturally responsive interventions that can enhance community well-being and support those affected by trauma.

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Appendix A.

Consent Form

Research Title: Coping and Growth through Adversity in Belizeans

We invite you to take part in our research study. Please take your time to read the following information. Feel free to ask if you need more information or if there is anything that you do not understand. To participate in this study, you must be: Belizean, over 18 years old, willing to discuss your experiences, and fluent in English.

What is the purpose of this study?

We want to understand the challenges you have faced and the ways you have found to overcome or grow through them.

Who is doing this research?

The research and interviews will be conducted by Joy Lee-Shi, a graduate student at Simon Fraser University (SFU). If you have any questions, please email her at xxx@sfu.ca. Joy is supervised by Dr. Robert Ley at SFU. You can contact him at ley@sfu.ca.

Do I have to join this study?

It is completely your choice. If you decide to join but change your mind later, that's okay. You can stop the interview at any time, even in the middle or end. You can ask for your data to be removed from the study and destroyed within one month after your interview.

What will happen if I join this study?

You will be asked to take part in an interview about your experiences with coping and growth through adversity. The interview will last about one hour or as long as you would like to talk about your experiences. With your permission, the interview will be audio-recorded in Zoom. You can stop the interview at any time, and you do not have to answer any questions you don't want to. Later, we will transcribe then delete the recording.

Where and when will the interview take place?

The interview will take place on Zoom. We will arrange a date and time that works best for you.

Are there any risks in taking part?

We do not think there is anything in this study that could harm you. However, emotional discomfort might happen during the interview. You can stop the interview anytime without giving a reason. Helpful mental health resources will be provided after the interview. If you agree to join, you can still take legal action if you are harmed by the research.

Are there any benefits in taking part?

You will help us develop a better understanding of how Belizeans cope and grow through adversity. This knowledge can help improve support services and inform future research in Belize.

What about my privacy?

Anything you share with Joy will be kept strictly confidential.

Your interview will be recorded and labeled with a code, not your name. We will use Otter AI to transcribe the recordings. This data will be encrypted and only Joy can access it. While it's being transcribed, your data will be temporarily stored on Otter servers in America. After the transcription, your data will be deleted from Otter servers.

Any identifying details like place names and people's names will be removed from the transcripts. The audio recordings will be destroyed once they are transcribed, checked, and anonymized. We may use quotes from the interviews in our study report, but no one will be able to identify you from these quotes.

While the study is ongoing, all research data will be encrypted and stored on a password-protected computer. Encrypted backups will be stored on OneDrive servers at Simon Fraser University in Canada. At the end of the study, the data will be kept on a password-protected computer for up to 5 years.

What will happen to the results of the study?

Anonymized data may be incorporated into future research conducted by Joy. The results will be written up for a course and/or thesis report. Research findings may also be published in an academic journal or presented at conferences. If you would like to receive a copy of the findings, please let us know by using the contact information provided, and we will happily provide you with one.

Who can I talk to if I have questions?

If you have any questions, you can email or direct message Joy at xxx@sfu.ca. You can also email the research supervisor, Dr. Robert G. Ley, at Simon Fraser University at ley@sfu.ca.

What if I am unhappy or if there is a problem?

If you have any concerns about your rights as a research participant and/or your experiences while participating in this study, please contact the Director, SFU Office of Research Ethics, at dore@sfu.ca or 778-782-6593.

Who is funding the research?

This study is not receiving any external funding or financial support. The research is solely funded by the primary researcher, Joy.

What will I receive for participating in this study?

As a thank you for your time and contributions, you will receive \$20 BZD after the interview. This will be given to you even if you decide to stop the interview early or choose not to answer certain questions.

How do I agree to join this study?

You can schedule a time and date for a Zoom interview with Joy. During the interview, Joy will ask you if you have read this consent form and if you consent to participate in the study. If you have any questions before the interview, feel free to ask.

[Thank you for taking the time to read this document!](#)

Appendix B.

Semi-Structured Interview & Distress Protocols

This protocol aims to gather comprehensive and culturally relevant insights into how Belizeans experience coping and growth through adverse experiences.

Informed Consent

A participant information sheet will be emailed to the participant before the interview. If the individual agrees to participate in this study, the researcher will schedule an interview. Before the interview, allocate a unique identifier to the interviewee. During the beginning of the interview, the interviewer will verbally ask for consent for research participation and audio-recording the interview. If the participant consents to both, the interviewer will start the audio-recording, and ask for consent again, for documentation.

Introduction Script (~5-10 min)

“Hello [interviewee’s name]. My name is Joy, and I am conducting a study on coping and growth through adversity in Belizeans. The purpose of our study is to understand how people in Belize experience and cope with adversity.

Thank you for agreeing to this interview. Before we proceed, I would like to explain a few important details about the interview and obtain your consent. This interview will involve answering a series of questions regarding your background and experiences. It should take about one hour or as long as you would like to talk about your experiences.

Please know that your responses will be kept completely confidential. We will not use your name or any identifying information in any reports or publications from this study. You can skip any questions you do not feel comfortable answering and can stop the interview at any time without any consequence or explanation.

If you agree to be recorded, the recordings will be stored securely and will be destroyed after they have been transcribed and analyzed.

Before we start, I want to ask if you have read the participant information sheet that I sent to you. Do you have any questions about the study or the information provided in the sheet?

[Pause for questions]

I also want to clarify that I am not a licensed or professional therapist; I am just a researcher. The focus of our discussion will be on understanding your experiences and perspectives.

Could you please let me know your preferences regarding participation and audio recording?”

Would you like to participate in this interview?

- Verbal Consent was obtained from the study participant
- Verbal Consent was NOT obtained from the study participant

Can we audio-record this interview?

- Verbal Consent was obtained
- Verbal Consent was NOT obtained

If the interviewee consents to participation and being audio-recorded, jump to Script A. If the interviewee consents to participation, but not to being audio-recorded, jump to Script B. If the interviewee does not consent to participation or audio-recording, jump to script C.

Script A: Interviewee Consents to Participation and Being Audio-Recorded

"Thank you! I will start the audio recording now.

[Start audio recording]

"Let's begin our interview for the study on coping through adversity. For the record, I would like to confirm your consent for the interview and audio recording.

Do you consent to participate in this interview?

[Wait for response].

And do you consent to this interview being audio-recorded?

[Wait for response].

Thank you for your consent. Before we begin, do you have any questions about the interview or the study?

[Pause for any questions.]

Let's proceed with the interview."

Start Interview

Script B: Interviewee Consents to Participation but Not to Being Audio-Recorded

"I understand and respect your decision. To ensure accuracy in capturing your responses, I would like to take detailed notes of our conversation. This would be done during the interview, and then from memory after our conversation. Do you consent to this?

[Wait for response]."

If the interviewee consents to data collection through interviewer note taking, jump to Script B1. Else, jump to Script B2.

Script B1

"Thank you for your consent. Before we begin, do you have any questions about the interview or the study?"

[Pause for any questions.]

Let's proceed with the interview."

Start Interview

Script B2

"I understand that you do not consent to any form of data collection, which includes audio recording, detailed note-taking, or notes taken from memory.

For the purposes of this research, it is important to accurately document our conversation to ensure that your insights can be accurately represented and analyzed. Without your consent to data collection, we won't be able to proceed with the interview as part of the study.

If you have any questions or if you reconsider in the future, please feel free to contact me. Thank you for your time.

However, if you are interested, we can have an informal discussion about the topic. This conversation would not be part of the official research data and would not be documented in any way. Would you be interested in having an informal discussion instead?

[Wait for response]."

Script C: Interviewee Does Not Consent to Participation and Audio-Recording

"I understand that you do not wish to participate in this study. Your decision is completely respected, and there are no consequences for choosing not to participate.

If you have any questions or if you reconsider in the future, please feel free to contact me. Thank you for your time.

If you are interested, we can have an informal discussion about the topic. This conversation would not be part of the official research data and would not be documented in any way. Would you be interested in having an informal discussion instead?

[Wait for response]."

Interview

Section 1: Background Questions (~5-15 minutes)

- Can you please tell me your age?
- What languages or dialects do you speak fluently?
- Can you tell me about your ethnic and cultural background?
- Can you describe your religious or spiritual beliefs and practices?

- Which region do you currently live in?
- Where do you currently work?
- Tell me a little about your work and what you do?
- What gender do you identify as?
- How would you describe your sexuality?

Section 2: Belizean Experiences with Trauma and Adversity

1. What are some very psychologically stressful events that you have experienced?

Follow-up Questions

1. How did you deal with this experience?
2. What sort of support was available?
3. Do you think you've developed as a person after this experience?
4. What noticeable changes have occurred in you following this experience?
5. What do you think caused these changes?

Probes

1. Tell me more about it?
2. Could you elaborate?

Closing Questions

1. Is there anything more you would like to tell me about what you have said?
2. What questions do you think I should have asked?
3. What questions do you have for me?

Closing Script

"Thank you very much for taking the time to speak with me today and for providing me with this information. It has been interesting and very helpful in my research. Also it may help others in coping with challenging life events.

Next, I will be analyzing the information gathered from these interviews to identify common themes and insights.

Please rest assured that all the information you have shared will be kept completely confidential. Your responses will be anonymized, and no identifying information will be used in any reports or publications resulting from this study.

Before we wrap up, do you have any final questions or comments about the interview or the study?

[Pause for any questions.]

I will email you a copy of a debriefing document that provides more information about the study and your rights as a participant. If you have any further questions or concerns after reading these documents, please do not hesitate to contact me.

If you have any questions or further thoughts after this interview, please feel free to contact me through email or direct messaging.

Once again, thank you for your time and contribution.”

Distress Protocol

Distress Protocol 1: Managing distress in an interview

1. Recognizing Distress

Through behaviour intervention work, the interviewer (JLS, co-investigator) is familiar with recognizing signs of distress. However, this experience is specific to children with neurodevelopmental conditions such as ASD and ADHD. JLS also has extensive experience in teaching a wide range of ages (childhood to late adulthood) as a TA at SFU and as a music teacher, where recognizing distress is an essential and often utilized skill.

2. Stage 1 Response:

- Stop the interview
- Check in with the participant: “I noticed that you seemed uncomfortable with the last question. Would you like to take a break, skip that question, or talk about something else?”

3. Review

- If the participant feels ABLE to carry on, resume the interview.
- If the participant is UNABLE to carry on, proceed to the next stage (Stage 2 Response)

4. Stage 2 Response:

- Terminate the interview: “Would you like to end the interview here?” If a participant chooses to end the interview due to distress, this would be accepted without question.
- Participants will be emailed a debriefing form. In the debriefing form, participants will be provided with information about resources for emotional support.

Appendix C.

Debriefing Form

Research Title: Coping and Growth through Adversity in Belizeans

Belize faces significant challenges with high rates of violence. With a population of 397k, reported domestic violence occurs at 304 per 100k individuals, sexual violence at 20.5 per 100k, and ten homicides monthly (Belize Crime Observatory, 2023; Statistical Institute of Belize, 2024). 77.1% of youths have experienced more than 10 traumatic incidences (Gayle et al., 2010), and 49% of gang-involved individuals report daily shootings in their neighborhoods (Young, 2019). However, there is little research on trauma and none on post-traumatic growth (PTG) in Belize.

Existing gang prevention and intervention services exist, their foundation in evidence-based research, effectiveness, and sustainability are widely criticized (Young, 2019). Thus, through semi-structured interviews, we seek to develop a comprehensive understanding of how Belizeans cope and grow from adverse experiences, aiming to inform future research and interventions for resilience and recovery.

Emotional Well-being and Support:

In our interview, we discussed topics related to resilience and adversity, which may bring up strong emotions. We encourage you to reach out to mental health professionals if you feel distressed.

Counseling and Health Centers

1. Community Counseling Center (free) - cs.coordinator.crd@humbandev.gov.bz
 - a. Walk into the HUB on Chetumal St, Belize City, OR call 227-1406
 - b. Bachelor and master level practitioners. Bachelor level practitioners are presently pursuing master level degrees and are closely supervised by master level practitioners through peer supervision.
2. Health Center Ladyville – 225-2764
3. Health Center Hatteville – 225-6106
4. Cleopatra White Polyclinic – 223-5213 OR 223-0131 OR 223-4885
5. Matron Roberts Polyclinic – 227-7170 OR 227-7176
6. Port Loyola Health Center – 227-5354

Online Directory of Mental Health Professionals in Belize

<https://www.mindhealthconnect.com/professionals-in-the-field/>

Confidentiality:

Your interview data will be absolutely confidential. Your name will be replaced with a code number, and only Joy will see your raw data.

Contact Information:

Your participation in this study is greatly appreciated. If you're interested in obtaining a copy of the results once the study is complete, you can contact the primary researcher, Joy, at xxx@sfu.ca. If you have a more general interest in this area of research, you can also contact Joy to learn more.

Complaints and Concerns:

Any ethical concerns about the study may be directed to the SFU Office of Research Ethics at dore@sfu.ca or 1-778-782-6618. When contacting the Ethics Committee, please provide the name or a description of the study, the researcher(s) involved, and the details of the complaint you wish to make.

Thank you so much for your time and contributions!!